दिल्ली प्रौद्योगिकी विश्वविद्यालय

DELHI TECHNOLOGICAL UNIVERSITY

Established by Govt. of Delhi vide Act 6 of 2009

(Formerly Delhi College of Engineering)

स्वच्छ भारत

एक कदम स्वच्छता की ओर

Prof. Prateek Sharma
Vice-Chancellor

Message

It gives me immense pleasure to announce that Delhi Technological University is commencing Ph.D. admissions for the Session: August, 2026.

Delhi Technological University is globally known for outstanding education, research and innovations. The University currently offers various interdisciplinary and industry relevant programmes in science, technology, management and allied areas at undergraduate, post- graduate and doctoral levels.

Students admitted to DTU, through their dedication, discipline and steadfastness can go on to become professionals and impactful leaders. DTU provides them with an environment to shape their talent as DTU ensures that every step of a student's journey is designed keeping in mind holistic development. This is coupled with a diverse range of extra-curricular activities throughout the year, which help students develop various skills to facilitate them throughout their lives.

Over the years, DTU has established itself as the University of unshakable repute. Hence, getting admission to DTU has reached great heights on the national and international stages and continues to make us proud. The Conjoined efforts of relentless students, faculty, administration and the staff have preserved an exceptional environment in DTU that allows persistent exchange of information and upholds the unmatched excellence associated with this University for eight decades.

We aim to nurture the students holistically and endeavour to foster a culture in which virtues and skills are instilled in them, along with a concern for society and its well-being.

I send my best wishes to the candidates applying for admission to the Delhi Technological University.

(Prof. Prateek Sharma)

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About DELHI TECHNOLOGICAL UNIVERSITY

Delhi Technological University has been a beacon of academic excellence and innovation for the past 85 years. The university has garnered recognition for being an eminent institution, known for its academic research and entrepreneurial excellence in many fields. Delhi Technological University has adopted the principles of knowledge, critical thinking, creativity, and innovation at the core of its education policy, all while embracing integration, access, and inclusivity. With an aim to provide a holistic education, conventional classroom teaching has been merged with innovative methods, fostering curiosity and academic prowess in our students. From its inception as Delhi Polytechnic to its present-day identity as Delhi Technological University, the university has always championed the cause of providing students with a high-quality education that allows them to flourish in their chosen domains. Along with imparting knowledge, the University has achieved pioneering advances in undertaking creative and internationally relevant research and innovation over the years.

With 85 years of its long-established tradition of excellence in engineering & technology education, research and innovations, Delhi College of Engineering, (initially established with the name – Delhi Polytechnic) came into existence in the year 1941 to cater the needs of Indian industries for trained technical manpower with practical experience and sound theoretical knowledge. The institution was set up at the historic Kashmere Gate campus as a follow-up to the Wood and Abott Committee of 1938. It comprised a multi-disciplinary and multi-level institution offering wide-ranging programmes in engineering, technology, arts and sculpture, architecture, pharmacy, and commerce. In 1952, the college was affiliated with the University of Delhi and started formal degree-level Programmes. The department of Architecture later became the School of Planning and Architecture (SPA), now a Deemed University and Institution of National importance. The department of Arts and Sculpture became the College of Arts, and the departments of Chemical Technology and Textile Technology were shifted to mark the beginning of the IIT Delhi at its new campus at Hauz Khas. Further expanding its academic reach, the Department of Pharmacy evolved into the College of Pharmacy under the Directorate of Training and Technical Education, Delhi. The Department of Commerce was later abolished, and the Faculty of Management Studies of the University of Delhi was established. Delhi College of Engineering is thus the mother institution of several national projects, including IITD, SPA, College of Arts, and even the famous FMS.

After the reconstitution of the Delhi College of Engineering by the Government of NCT of Delhi in 2009, by Act 6 of 2009, passed by the assembly of the NCT of Delhi, DCE became Delhi Technological University. It is a non-affiliating, teaching, and research University, committed to achieving excellence in Engineering, Science, Technology, Management, and allied areas. Over its rich history of over 85 years, the institution has been serving the nation and the global community since its inception in 1941, by providing trained manpower of the highest quality in the field of engineering and technology, and is globally well known for its outstanding education, research, and innovations.

DTU remains committed to cultivating a culture where scientific research fuels technological advancement and technological ingenuity enriches scientific exploration so that science and engineering seamlessly converge. The University is dedicated to nurturing innovation by promoting industrially relevant projects, technology incubation, and research-driven education with the help of a dedicated Incubation and Innovation Foundation. With a strategic focus on cutting-edge research and development, DTU continues to expand its footprint in pioneering innovations, interdisciplinary collaboration, and entrepreneurial ventures.

Upholding and uplifting its long and remarkable history in education and research, the University currently offers various inter-disciplinary and industry-relevant programs in Science, Engineering, Management & Allied areas at the undergraduate, postgraduate and doctoral level. The University has established a strong academia-industry interface and has collaborations with reputed research organisations, industries, and premier institutions. Many alumni of the institute have achieved remarkable success both in India and abroad, bringing honour to the name of the University. Their contributions reflect the strong foundation and values instilled during their time at the University. DTU places great emphasis on nurturing national character, self-confidence, and leadership among its students and fosters an ecosystem that encourages creativity, imagination, and innovation. The University is committed to developing not only skilled professionals but also responsible, forward-thinking leaders. With its rich legacy and progressive outlook, DTU continues to empower students to make a lasting impact and stay at the forefront of global academic excellence.

VISION

To be a world class university through education, innovation and research for the service of humanity.

MISSION

To establish centres of excellence in emerging areas of science, engineering, technology, management and allied areas.

To foster an ecosystem for incubation, product development, transfer of technology and entrepreneurship.

To create environment of collaboration, experimentation, imagination and creativity.

To develop human potential with analytical abilities, ethics and integrity.

To provide environment friendly, reasonable and sustainable solutions for local & global needs.

LOCATION

Delhi Technological University is situated at Shahbad Daulatpur, Rohini in North - West Delhi, India. It is approximately 32 kilometers from the Indira Gandhi International Airport, New Delhi and the nearest Metro stations are Samaypur Badli/Rithala. Once at Samaypur Badli/Rithala, board local transport, auto or bus to get down at DTU, which is 3-4 kms far from Samaypur Badli/ Rithala Metro Station.

PROGRAMMES OFFERED

The university offers 17 Undergraduate Engineering Programmes, 25 Postgraduate Engineering Programmes, 11 M.Tech by Research, 5 MBA Programmes, 03 M.Des Programme, 5 M.Sc. Programmes, 05 Integrated B.Sc. – M.Sc. Programmes, and 01 M.A. Economics Programme. Besides this, the University offers other 3 undergraduate programmes in the field of Design (B. Des.), Management (BBA) and BA (Hons) Economics. The university offers 28 Ph.D Programmes in all areas of engineering, science, management, English and economics. The UG and PG programmes of DTU provide most modern curricula, based on the Choice Based Credit System (CBCS), having rich mix of courses from science, engineering, management, social sciences, humanities, fine arts, liberal arts, classical music, sports, etc. The course curricula have been developed with a view to integrate advancements in science and engineering, thereby incorporating industry relevant technologies. To offer further flexibility, there is provision for credit transfer and earning credits through massive online courses (MOOCs) from different platforms such as NPTEL, SWAYAM, etc. The curriculum is regularly updated keeping in view the new technologies and changes in industries and society.

RANKING AND REWARDS

The university is having ISO 9001:2015 certification since 27.11.2018, accredited with 'A' grade by NAAC (National Assessment and Accreditation Council) and has been accorded 2(f) and 12-B status by the University Grants Commission (UGC). Many of its UG & PG engineering programmes are also accredited by the National Board of Accreditation (NBA). The University is consistently ranked among best 10 engineering institutions as per the various independent surveys on best engineering institutions of the country. The university has been ranked 9th by India Today's Top engineering colleges in India 2025. The 2025 NIRF rankings placed DTU at the 30th position among the engineering institutions and has been ranked among top 10 institutes in India in terms of placement and higher studies as per NIRF data in the Engineering category. It has been ranked 7th in the university category and 9th among government engineering colleges by India Today in 2025. DTU has been placed at 201-250 bracket in the Asia University Ranking, 601-800 bracket in the Engineering & Technology Ranking and 801-1000 bracket in the World University Ranking published by Time Higher Education in 2025.

CAMPUS AND INFRASTRUCTURE

DTU has around 164 acres of a lush green, tech- savvy main campus, consisting of 16 academic departments, research centers, and residences for students, faculty, and staff. At present the university has around 15,000 students in its undergraduate, postgraduate, and Ph.D. programs. DTU has an EDUSAT Studio utilized for recording of lectures, events, and talks. Besides the main campus, the university has another campus in East Delhi, where the Executive MBA (Data Science and Analytics), MA (Economics), BA (Hons.) Economics and BBA (Hons.) Programmes are offered. The newly established East Delhi Campus of DTU has been functional since the 2017-18 academic session. It is located at Vivek Vihar, Phase II, Delhi. This campus endeavours to provide quality education, research, and innovation in the emerging areas of management, relevant to industry and society.

COMPUTER CENTRE

DTU has a well-equipped centralized computer centre to cater to the needs of students and faculty in the university. It is housed, in a magnificent state-of-the-art building having specialized laboratories to provide variety of platforms and computing environment for UG, PG and research students. The centre possesses a number of servers and over 275 Dell intel core i5 computer systems. In addition, the centre has more than 15 servers hosting different applications such as websites & portals, SPSS, Mathematica, MatLab, DNS, LDAP, proxy, Email services, Network Monitoring System (NMS) etc. and 4 SUN CAD workstations meant for use by UG/PG/PhD students for their projects and research work. The centre is fully networked through high-end intelligent Juniper/Avaya/CISCO/ Brocade/ Ruckus manageable switches, and possesses round the clock two leased lines of 10 Gbps link of NKN and 1Gbps link of Reliance Jio with shared bandwidth in different pipes for the Wi-Fi connectivity in the Library, Academic Departments, Administrative Blocks, Sports Complex, Faculty Residence and Hostel blocks of the campus, with internet facilities on all the nodes. It also has the latest versions of compilers, scientific, technical and engineering software, training kits etc. for the students of different branches of engineering.

CENTRAL LIBRARY

Delhi Technological University's Central Library is a beacon of knowledge, innovation, and intellectual growth at this historic institution. Established in 1941, the DTU Library has evolved to meet the changing needs of students, faculty, and researchers, blending tradition with modern advancements in technology. It is an ISO 9001-certified Library.

Expansive Collection, Dedicated Spaces, and Advanced Facilities

Spanning 5,000 square meters, DTU Library offers over 230,000 books that span engineering, technology, design, management, economics, literature, and sciences. Library resources support both academic and professional growth, with materials available in both Hindi and English to serve our diverse academic community.

The library provides seating for 500 students, ensuring comfort and accessibility, with campus-wide Wi-Fi connectivity enabling uninterrupted digital access. The library is fully computerized, with all collections barcoded for efficient management. Online Public Access Catalog (OPAC) is accessible over the internet, allowing users to search for books, make suggestions for new acquisitions, and even reissue borrowed books from anywhere, making it easier than ever to engage with library resources remotely.

Additionally, the Library Conference Room, with a capacity of 100, serves as a versatile space for workshops, seminars, and presentations. Equipped with advanced audio-visual technology, it is regularly used for faculty development sessions, guest lectures, and research symposia, enhancing the library's role as a collaborative and engaging learning environment.

Inclusive Membership and Family Engagement

DTU Library extends its resources beyond students and faculty, offering membership to the family members of faculty and staff to foster a culture of continuous learning.

Embracing Digital Transformation

Committed to digital accessibility, DTU Library actively participates in the National Digital Library of India (NDLI). Through the

Million Books Project, we have digitized 400 rare books, making these valuable resources accessible to the nation. Our digital collection also includes a collection of selected manuscripts in digital form and multimedia resources. The library provides access to 24 academic databases, numerous e-books, and a remote access facility that enables students and faculty to access resources anytime, anywhere.

To support academic excellence, we provide tools for plagiarism detection, grammar improvement, and access to industry standards from BIS, ASTM, and IEEE, as well as case studies from Harvard Business Publishing. Digital services not only ensure high-quality resources but also uphold integrity in research and academic writing.

Information Literacy Programs: Empowering Knowledge

Information Literacy Programs aim to equip students, faculty, and staff with essential skills to navigate, evaluate, and apply information in their studies and research. Key programs include:

1. Library Orientation: Introduces new students to library resources, including OPAC and online search functionalities.

2. Database and E-Resource Training: Workshops on how to effectively search databases, use the OPAC, and access e-resources remotely.

3. Academic Integrity and Plagiarism Awareness: Training sessions on ethical research practices, citation, and using plagiarism detection software.

4. Research Skills Development: Advanced workshops for postgraduate and PhD students on literature review, academic writing, and data analysis.

5. Digital Literacy & Faculty Training: Sessions for faculty on digital tools, remote access to resources, and customized support for research and teaching needs.

Information Literacy Week is a highlight event that introduces new themes each year, focusing on crucial topics in research, publishing, and digital scholarship.

Digital Archive and Research Support

Digital archive preserves DTU's academic contributions, including PhD theses, master's dissertations, and faculty publications. Current awareness bulletins keep the DTU community informed of recent publications by students and faculty, as well as new library acquisitions, fostering a collaborative and informed academic environment.

Interlibrary Collaboration and Networked Resources

As a member of INFLIBNET and DELNET networks, DTU Library enhances resource-sharing capabilities, offering students and faculty access to a broader network of materials across institutions and promoting interlibrary collaboration.

At DTU Library, each book, digital resource, and program opens a world of opportunity. Dedicated to supporting the academic and professional journeys of our students, faculty, and alumni. DTU Library is a place where every corner holds the promise of discovery and excellence.

HOSTEL

Hostel life is one of the most enjoyable and memorable time of one's life. There are eleven boys' hostels and three girl's hostels in DTU, besides, one separate hostel for international students (boys). Each hostel in the campus gives each individual ample opportunity to develop various qualities as each hostel is equipped with recreation room, reading room, mess and gymnasium. Additionally, every hostel subscribes to the latest magazines and newspapers for the residents. The hostels are connected to the campus via the campus wide wi-fi network and LAN which enables the residents to browse the internet and access the online library resources for their academic and research related work. The information of all available accommodation will be posted on the University website. However, limited seats could be provided inside the University premises. In addition, the mess facility at the University can be availed by all the students.

CENTRE FOR EXTENSION & FIELD OUTREACH

Centre for Extension and Field Outreach was established in DTU in the year 2018. The various activities/ program performed by the Centre is to sensitize the students to develop social values, widespread their responsibilities and knowledge in societal issues and problems by making them to involve with the community people. DTU is a Participating Institute under under "Unnat Bharat Abhyan"- a Project of Ministry of HRD, Govt. of India and adopted five villages and are conducting classes in their schools. Directorate of Education, Govt. of NCT of Delhi awarded a Project "Youth for Education" and has launched "Desh Ke Mentor", which is one of the largest mentoring program in school education. Centre has also started a certificate course titled as "Basic Computer Course". Under Lab on Wheels (LOW) Scheme for the candidates from the Government Schools of NCT of Delhi or from society. Centre at DTU is coordinating with Delhi Police conducting Skill development program through one-month basic computer training to Juveniles in conflict with law/ weaker section in Rohini. Centre is regularly organizing Seminars/ online webinars/ workshops/ Awareness programs etc., and is working towards increasing productivity, enhancing skills and abilities, focusing on growth and helping people to work on their own future development. Additionally, NSS at DTU facilitates enhanced student engagement with community contributing to deeper reservoirs of ideas.

INNOVATION AND INCUBATION FOUNDATION (DTU-IIF)

DTU-IIF is a Technology Business Incubator (TBI) established in 2016 as a non-profit section 8 company. Currently, this TBI is supported by the Government of Delhi and Delhi Technological University. DTU-IIF helps start-up companies and individual entrepreneurs to develop their business ideas by providing a range of services including co-working office space, mentoring support, funding support with venture capital financing, and other supports & resources they need, all under one roof. During last five years, IIF provided 70 lakhs of funds to 56 start-up companies. Also, DTU-IIF promotes the culture of innovation and Entrepreneurship by organizing various webinars/workshops/Hackathons, etc. The Business Review Committee screens the new ideas and recommends incubation at DTU-IIF. The Finance Review Committee recommends the investment of Rs. 7.5 lakh per start-up. Also, DTU-IIF provides pre-seed support of Rs. 50,000/- to develop ideas.

SPORTS AND OTHER OUTDOOR ACTIVITIES

The students of DTU are provided with excellent facilities for indoor and outdoor games. DTU has 4 x 400 m racing track, fields for football, hockey, cricket, courts for volleyball, basketball, tennis, badminton, along with facilities for indoor games. A well-equipped gymnasium is also available in the campus in addition to gym facilities in each hostel. The university has appointed coaches in almost all the games to coach the students and prepare university teams. Students are encouraged to participate in various sporting events and tournaments held in, and around, NCR of Delhi. From academic year 2018-19, as per the revised curriculum, the university offers foundation electives to the students of first year and second year and in these sports have big share of electives.

A large number of bright and capable scholars, having graduated from the Institute, have distinguished themselves by means of their extraordinary achievements in their chosen professions and by their contributions to the society at large.

DCE-DTU ALUMNI NETWORK

DCE-DTU Alumni are serving leadership positions in many of the best-known companies in India and abroad, in marketing, finance, human resources, information technology, research & analytics, innovation & entrepreneurship. And the worldwide network of illustrious alumni includes world-known personalities like Prof. Vinod Dham (Father of the Pentium Chip), Dr. Raj Soin (Founder, CEO of Soin, and LLC), Prof. D. Yogi. Goswami (Inventor, Author, Entrepreneur and Educator), Dr. Durga Das Aggarwal (President, CEO Piping Technology & Products, Inc). Mr. Vijay Shekhar Sharma, (Founder of Paytm), Sh. Karnal Singh (Former Chief of Enforcement Directorate), Sh. Arun Goyal (Member-CERC & Former Secretary, Cabinet Secretariat).

Alumni have been traditionally contributing generously towards placement, opportunities, sponsorships/ Fellowship programs and infrastructural developments of their alma mater. Donations for Raj Soin Hall by Dr. Soin, Clean Energy Research Centre establishment by Prof. Yogi Goswami, and several scholarships for the students of DTU have shown the dedication of the alumni for the betterment of their alma mater. Vinod Dham has sponsored "Centre of Excellence for Semi-conductors and Micro-electronics" to establish centralized state of art infrastructure facility for device design / material research / fabrication for cutting edge R&D in Semi- conductors and Micro-electronics.

EVENTS AND FESTIVALS

The university organizes annual cultural, literary, sports and technical festivals. These festivals not just provide an opportunity to the students to connect with the professional world, but also display their creative and technical skills in several interesting events and activities organized during the fests. The ENGIFEST, one of the most well attended student's cultural event in northern India and the YUVAAN, the literary Fest, is annual cultural extravaganza of the university and offers a good mix of literary, cultural, and entertainment events. The INVICTUS is annual technical festival of the university where all technical societies of the university host various technical activities and competition. The AAHVAAN is the annual sports fest organized by DTU sports council.

MEDICAL FACILITIES

DTU has a well-equipped health care centre. The medical practitioners are available to the students requiring medical attention. The healthcare centre has specialized medical practitioners including ENT, dental care, Physiotherapy, Nutrition, Gynaecology and Obstetrics etc. Further, medical camps are also being organized by the University on regular basis. In addition, Ambulance facility is also available in case of emergency. The University has also tie-ups with the major hospitals of Delhi for emergency cases.

More information about DTU can be accessed at **www.dtu.ac.in.**

Bank and Post office

The State Bank of India has a branch with ATM facilities operating from the Institute campus. There is a Sub-Post Office in the name of "DCE" in the Institute campus and its PIN Code is 110042.

Guest House

The University has a guest house intended to accommodate various official visitors and participants of seminars/Workshops/ Conferences,/Training programs organized by the University/Centres/Departments. This well-equipped guest house features eight double-bedded rooms, each with modern facilities to ensure a comfortable stay for guests.

1. RESEARCH FACILITIES AT DTU

All the academic departments of the university have well equipped research laboratories and workshop facilities. In addition, there are a number of central facilities such as Central Workshop, Solar Energy Centre, Central Instrumentation, Centres for Advanced Studies & Research in Automotive Engineering, TIFAC-CORE, Central Library and Computer Centre. The Central Library has more than 2,30,000 books, a large collection of back volumes of periodicals, standard specifications and other literature. It subscribes more than 83,505 current journals in Science, Engineering, Humanities and Social Sciences as e-resources. DTU has a well-equipped centralized Computer Centre which provides state of art high- end networked computing facilities to students and staff.

The University has many research collaborations with leading universities and Institutes in Korea, Singapore, France, Florida USA, Africa and China. As part of these collaborations, the students get opportunities to carry out joint research projects with faculty and students from these institutions.

The location of DTU in close proximity to several leading R&D Centres namely NPL, INMAS, FICCI, CSIR, etc. and other major industrial establishments which offers excellent opportunities to interact and plan research programmes and projects in collaboration.

2. FACULTY AND RESEARCH

The university has a very talented pool of experienced, as well as young faculty members who are well qualified in their area of specialization and have very good national and international exposure. To engage the students and faculty in research and innovation the university offers provisions like funding for students' innovative projects, financial assistance to students for attending internship overseas, research project grants to all faculty members, etc.

To celebrate the individual's excellence in research, the university gives Research Excellence Awards to researchers in three categories of awards annually, namely, Outstanding Research Awards, Premier Research Awards, and Commendable Research Awards. The awards are open to all the researchers of DTU. The University provides funds to faculty and students to organize and attend various faculty development programs, seminars, and conferences.

3. Ph.D. PROGRAMME

The University is inspired by talent and driven by innovations and is firmly committed to provide industry-relevant, socially-responsible manpower to meet the challenges of 21st Century. The vibrant culture of research and innovations in DTU campus inspires students from UG level onwards to engage in cutting edge technology development and discover the value and worth of the knowledge acquired by them during their studies.

The University offers Ph.D. programme in a wide range of areas in Engineering, Science, Management and Humanities. The academic programme leading to the Ph.D. degree is broad-based and involves a course credit requirement and a research publications leading to the submission. Facilities for research work leading to the Ph.D. degree are available in Departments of Mechanical Engineering, Information Technology, Environmental Science and Engineering, Electrical Engineering, Electronics & Communication Engineering, Humanities & USME, Design, Delhi School of Management, Computer Science & Engineering, Civil Engineering, Biotechnology, Applied Physics, Applied Mathematics, Applied Chemistry, Software

Engineering, Centre of Excellence for Electrical Vehicle and Related Technologies, Centre of Excellence for the Science of Happiness, Department of Geospatial Science and Technology and Centre for Community Development and Research, Vinod Dham Centre of Excellence for Semiconductors and Microelectronics and Nodal Centre of Excellence in Energy Transition.

DTU also offers Ph.D. Admissions under the AICTE QIP (Quality Improvement Program) and AICTE Doctoral Fellowship (ADF) scheme.

4. ADMISSION CATEGORIES

The applicant for admission to the Ph.D. programme shall be classified under any one of the following categories which will be decided and recommended by DRC.

4.1 Full-time Research student/ Candidate

Full Time University Research student/ candidate (With Fellowship or without Fellowship).

i. Full-Time students with DTU Fellowship admitted within the DTU Fellowship seat matrix.

   Further, if a candidate who is admitted as Full Time research scholar with DTU Fellowship chooses to withdraw from the Ph.D. programme, due to ANY reason, the candidate shall be liable to refund the full amount of fellowship disbursed to him/her.

ii. Full-Time students without Fellowship can be admitted over and above DTU Fellowship seat matrix. The number of such seats will be based on suitability of candidates and availability of slots with prospective supervisors.

iii. Full Time Sponsored Research student /candidates - Fully financed by the Govt./Semi-Government Organizations like UGC-NET, UGC-CSIR-NET, DBT, ICMR, IARI-JRF, ICCR/ MHRD/MEA etc., Government/ Private funded Research / Development Organization, Public Sector Undertaking, Educational Institution or a reputed industry etc. These seats with Fellowship from other agencies shall be over and above the seat matrix with University Fellowship.

iv. Full Time / Part Time Sponsored Research student/candidate nominated by the organization having MoU with the University, foreign students who apply through Ministry of Human Resource Development or under a Cultural Exchange Fellowship Programme by Government of India.

v. Direct admission to Ph.D. program for the DTU full-time B.Tech. (or) M.Tech. students is subject to fulfilling the guidelines laid down by the BOM in its 40th (agenda 40.5) (or) 37th meeting (agenda 37.7), respectively.

4.2 Part - Time Research student/ candidate

i. Research student / Candidate working in other organizations having MoU with Delhi Technological University.

ii. Candidate employed in Educational Institutions, R&D organizations and Government Department, Public Sector Undertaking and Candidates from industry/companies of high repute and a medium sized enterprise along with standing commitment to the exemplary standards namely ISO/ CMM or similar standard of respective area provided that the applicant possesses the minimum eligibility qualifications for the degree.

iii. Students with valid JRF may also be considered for admission in Part-Time Ph. D. Programme without fellowship.

4.3 Candidate/staff under R&D projects at DTU

Project staff under project sponsored by DST/UGC/ any government agency, industry or centres established from grant in aid from government or international agency at university.

4.4 Research student/Candidate working as a regular employee

Permanent academic staff of the Delhi Technological University (Including the academic staff of erstwhile Delhi College of Engineering).

4.5 Industry/ Working Professionals

DTU's Ph.D. Program for "Industry/ Working Professionals" is another step in the direction of encouraging innovation driven doctoral research for professionals of high standing and competence in their disciplines. Committed professionals drawn from Industries, R&D organizations and Government Departments will be able to utilise this opportunity to fulfil their aspirations of pursuing Ph.D. in their preferred areas of excellence. This Ph.D. program will play an integral role in establishing long lasting and fruitful ties between DTU and Industry professionals for pursuing high-value projects in the knowledge driven economy.

The Professional joining Ph.D. Programs at DTU will be able to update their knowledge and skills to grow and succeed in business environments. Exposure to relevant academic experiences and relationships will enhance the employability skills of the researchers. Further, Companies will be able to develop skilled human resources by training and supporting the next generation of researchers. Thus, their chances of thriving in the modern competitive market through innovation and knowledge exchange with university and research institutions will increase manifolds.

Guidelines for Ph.D. Program for Industry/ Working Professionals Essential Qualifications:

<table>
<tr><th>Broad Discipline</th><th>Eligibility criteria</th></tr>
<tr><td>Engineering/ Technology</td><td>1. Bachelor's / Master's in Engineering/Technology in relevant discipline or equivalent** degree with fulfilling work experience and CGPA given at point (3)</td></tr>
<tr><td>Science, Management, Humanities, Entrepreneurship</td><td>2. Master's degree in Sciences/Management/Humanities and Social Sciences in relevant discipline or equivalent** degree with fulfilling work experience and CGPA given at point (3).</td></tr>
<tr><td colspan="2">3. Work Experience and CGPA For all disciplines:
a) More than 05 years and less than 10 years work experience* CGPA of 7.0 on a 10-point scale or 70% marks
'OR'
b) More than 10 years and less than 15 years work experience* CGPA of 6.5 on a 10-point scale or 65% marks
'OR'
c) More than 15 years work experience* CGPA of 6.0 on a 10-point scale or 60 % marks</td></tr>
<tr><td colspan="2">4. Credentials of the company/ organization of the working professional applying for the program shall be assessed on the basis of the following mandatory criteria for all disciplines:
a) The reputation of the companies (private or government or PSU's), Research Organizations, Ministries of Central and State Governments or Union Territories or Recognized Research Institutes or Public Sector Undertaking or Semi-Govt. or Autonomous or Statutory organizations/ institutions or Registered Companies or industrial research and development organizations excluding academic institutions.
b) With standing commitment to the exemplary standard, namely, ISO/CMM or similar standard of respective areas mandatory for any enterprise/ company/industry/firm.
c) The candidate must have the working experience at least continuous 02 years in the respective organization/ institution at the time of application along with the work experience indicated in Point (3) above.
d) A research proposal approved by the prospective supervisor must be submitted by the candidate at the time of the application.</td></tr>
</table>

Work experience may include position in multiple organization(s), such candidate shall be working on industry oriented research problems.

***** Equivalence of degree will be decided by the University.***

i. **For all the broad disciplines:**

a) The candidate must meet the minimum eligibility criteria to be shortlisted for interview. In the absence of conversion of CGPA to percentage of marks (or vice versa) mentioned in the transcripts then the conversion formula of DTU will be applicable.

b) Candidates need to provide a 'NO OBJECTION' certificate issued from their company, stating it has no issues with the candidate pursuing Ph.D. under the proposed scheme "Ph.D. Program for Industry/Working Professionals".

ii. **Desirable Qualifications:**

Candidates having proven research capability and active participation record in devising/ designing, product development, planning, executing, analysing, quality control, innovating, training, technical books/research paper publications/ IPR/patents etc.

5. ADMISSION ELIGIBILITY

Academic Department	Disciplines Offered	Discipline Specific Eligibility Criteria
Applied Chemistry	Chemistry	Master's degree in Sciences in Chemistry / Applied Chemistry / Industrial Chemistry / Polymer Chemistry / Polymer Science / Electrochemistry / Pharmaceutical Chemistry / Material Chemistry / Material Science / Drug Chemistry / Medicinal Chemistry / Green Chemistry / Environment Chemistry / Environment Science / Chemical Science / Biochemistry / Nanomaterials / Nanoscience / Food Science / Metallurgy / Agrochemicals / and Chemistry related disciplines with a minimum 55% marks in aggregate or equivalent CGPA as determined by DTU.
	Chemical Engineering	Master's degree in Engineering/Technology in Chemical Engineering/ Chemical Technology / Polymer Engineering / Polymer Technology/ Textile Engineering / Textile Technology / Nanotechnology / Biotechnology / Biochemical Technology / Biochemical Engineering/ Bioprocess Engineering / Environmental Engineering / Food Technology and Chemical Engineering related disciplines with a minimum 55% marks in aggregate or equivalent CGPA as determined by DTU. OR Bachelor's degree in Engineering/Technology in Chemical Engineering/ Chemical Technology / Polymer Engineering / Polymer Technology / Textile Engineering / Textile Technology / Nanotechnology/ Biotechnology/ Biochemical technology / Biochemical Engineering/ Bioprocess Engineering / Environmental Engineering / Food Technology and Chemical Engineering related disciplines with a minimum 75% marks in aggregate or equivalent CGPA and having proven research capability.
Applied Physics	Physics	Master's degree in Engineering / Technology / Sciences in relevant disciplines or equivalent with a minimum 55% marks in aggregate or equivalent CGPA as determined by DTU. OR Bachelor's degree in Engineering/Technology in relevant discipline or equivalent with a minimum 75% marks in aggregate or equivalent CGPA and having proven research capability.
	Engineering Physics	Master's degree in Engineering/Technology in relevant disciplines or equivalent with a minimum 55% marks in aggregate or equivalent CGPA as determined by DTU. OR Bachelor's degree in Engineering/Technology in relevant discipline or equivalent with a minimum 75% marks in aggregate or equivalent CGPA and having proven research capability.

Academic Department	Disciplines Offered	Discipline Specific Eligibility Criteria
Applied Mathematics	Mathematics	Master's degree in Sciences / Arts in relevant disciplines or equivalent with a minimum 55% marks in aggregate or equivalent CGPA as determined by DTU.
	Mathematics and Computing	Bachelor's degree in Engineering / Technology and Master's degree in Engineering / Technology in relevant disciplines or equivalent with a minimum 55% marks in aggregate or equivalent CGPA as determined by DTU OR Bachelor's degree in Engineering/Technology in relevant discipline or equivalent with a minimum 75% in aggregate or equivalent CGPA as determined by DTU and having proven research capability NOTE: Entrance Exam will be 30% in Mathematics Domain and 70% in Computing Domain
Biotechnology	Biotechnology	Master's degree in Engineering/Technology/Sciences in relevant discipline or equivalent with a minimum 55% marks in aggregate or equivalent CGPA as determined by DTU. OR Bachelor's degree in Engineering/Technology relevant to Life Sciences with a minimum 75% marks in aggregate or equivalent CGPA and having proven research capability.
Civil Engineering	Civil Engineering	Master's degree in Engineering / Technology in relevant discipline or equivalent with a minimum 55% marks in aggregate or equivalent CGPA as determined by DTU. OR Bachelor's degree in Engineering / Technology in relevant discipline or equivalent with a minimum 75% marks in aggregate or equivalent CGPA and having proven research capability.
Computer Science & Engineering	Computer Science and Engineering	Bachelor's degree in Engineering/Technology and Master's degree in Engineering / Technology in Computer Science and Engineering/ Software Engineering / Information Technology/ Mathematics and Computing / Electronics and Communication Engineering or equivalent with a minimum 55% in aggregate or equivalent CGPA as determined by DTU OR Bachelor's degree in Sciences/Computer Applications and Master's degree in Computer Applications (with Mathematics at B.Sc./B.C.A level) with a minimum 75% in aggregate or equivalent CGPA as determined by DTU and having proven research capability
		OR Bachelor's degree in Engineering/Technology in Computer Science and Engineering/Software Engineering/Information Technology/ Mathematics and Computing/Electronics and Communication Engineering. or equivalent with a minimum 75% in aggregate or equivalent CGPA as determined by DTU and having proven research capability
Delhi School of Management	Management	Master's degree in Engineering / Technology / Sciences / Management/ Humanities/Commerce and Social Sciences in relevant discipline or equivalent with a minimum 55% marks in aggregate or equivalent CGPA as determined by DTU. OR Bachelor's degree in Engineering/Technology in relevant discipline or equivalent with a minimum 75% marks in aggregate or equivalent CGPA and having proven research capability.

<table>
<tr><th>Academic Department</th><th>Disciplines Offered</th><th>Discipline Specific Eligibility Criteria</th></tr>
<tr><td rowspan="2">Electrical Engineering</td><td>Electrical Engineering</td><td rowspan="2">Master’s degree in Engineering / Technology in relevant discipline or equivalent with a minimum 55% marks in aggregate or equivalent CGPA as determined by DTU
OR
Bachelor’s degree in Engineering / Technology in relevant discipline or equivalent with a minimum 75% marks in aggregate or equivalent CGPA and having proven research capability.</td></tr>
<tr><td>COE-EVRT</td></tr>
<tr><td>Electronics & Communication Engineering</td><td>Electronics and Communication Engineering</td><td>Bachelor’s degree in Engineering/Technology and Master’s degree in Engineering/Technology in in ECE/ Electrical and Electronics Engineering/ EE/ CSE/ SE/ IT /M & C or equivalent with a minimum 55% marks in aggregate or equivalent CGPA as determined by DTU
OR
Bachelor’s degree in Engineering/Technology in ECE/ Electrical and Electronics Engineering/ EE/ CSE/ SE/ IT /M & C or equivalent with a minimum 75% marks in aggregate or equivalent CGPA and having proven research capability.</td></tr>
<tr><td>Environmental Science and Engineering</td><td>Environmental Science and Engineering</td><td>Master’s degree in Engineering / Technology / Sciences / Management in the relevant discipline (Environmental Engineering / Environmental Science / Civil Engg. / Biotechnology / Chemical Engg. / other relevant branch) or equivalent with a minimum 55% marks in aggregate or equivalent CGPA as determined by DTU
OR
Bachelor’s degree in Engineering / Technology in relevant discipline or equivalent with a minimum 75% marks in aggregate or equivalent CGPA and having proven research capability</td></tr>
<tr><td rowspan="2">Humanities</td><td>English</td><td>Master’s degree in English or relevant branch or equivalent with a minimum 55% marks in aggregate or equivalent CGPA as determined by DTU.</td></tr>
<tr><td>Economics</td><td>Master’s degree in Economics or relevant branch or equivalent with a minimum 55% marks in aggregate or equivalent CGPA as determined by DTU.</td></tr>
<tr><td>Information Technology</td><td>Information Technology</td><td>Bachelor’s degree in Engineering/Technology and Master’s degree in Engineering / Technology in Computer Science and Engineering/ Software Engineering / Information Technology/ Mathematics and Computing / Electronics and Communication Engineering or equivalent with a minimum 55% in aggregate or equivalent CGPA as determined by DTU
OR
Bachelor’s degree in Sciences/Computer Applications and Master’s degree in Computer Applications (with Mathematics at B.Sc./B.C.A level) with a minimum 75% in aggregate or equivalent CGPA as determined by DTU and having proven research capability
OR
Bachelor’s degree in Engineering/Technology in Computer Science and Engineering/Software Engineering/Information Technology/ Mathematics and Computing/Electronics and Communication Engineering. or equivalent with a minimum 75% in aggregate or equivalent CGPA as determined by DTU and having proven research capability.</td></tr>
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<table>
<tr><th>Academic Department</th><th>Disciplines Offered</th><th>Discipline Specific Eligibility Criteria</th></tr>
<tr><td>Mechanical Engineering</td><td>Mechanical Engineering</td><td>Master's degree in Engineering/Technology or a Master's degree by Research in Engineering/Technology in Mechanical with specialization in Thermal/ Production / Design / Industrial Engineering having a minimum 55% marks in aggregate or equivalent CGPA as determined by DTU with Bachelor's degree in Engineering / Technology in Mechanical / Production / Production and Industrial/ Mechanical and Automation / Automobile Engineering or Equivalent
OR
Bachelor's degree in Engineering/Technology in Mechanical/ Production / Production and Industrial / Mechanical and Automation/ Automobile Engineering or equivalent having a minimum 75% marks in aggregate or equivalent CGPA and having proven research capability.</td></tr>
<tr><td rowspan="2">Software Engineering</td><td>Software Engineering</td><td rowspan="2">Bachelor's degree in Engineering/Technology and Master's degree in Engineering / Technology in Computer Science and Engineering / Software Engineering / Information Technology/ Mathematics and Computing / Electronics and Communication Engineering or equivalent with a minimum 55% in aggregate or equivalent CGPA as determined by DTU
OR
Bachelor's degree in Sciences/Computer Applications and Master's degree in Computer Applications (with Mathematics at B.Sc./B.C.A level) with a minimum 75% in aggregate or equivalent CGPA as determined by DTU and having proven research capability
OR
Bachelor's degree in Engineering/Technology in Computer Science and Engineering/Software Engineering/Information Technology/ Mathematics and Computing/Electronics and Communication Engineering. or equivalent with a minimum 75% in aggregate or equivalent CGPA as determined by DTU and having proven research capability</td></tr>
<tr><td>Computer Science</td></tr>
<tr><td rowspan="3">USME</td><td>Management</td><td>Master's degree in Management/Engineering/Technology/Commerce/ Economics and other behavioral sciences and allied relevant disciplines, or equivalent, with a minimum 55% marks in aggregate or equivalent CGPA as determined by DTU.
OR
Bachelor's degree in Engineering/Technology in relevant discipline or equivalent with a minimum 75% marks in aggregate or equivalent CGPA and having proven research capability.</td></tr>
<tr><td>Economics</td><td>Master's degree in Economics / Business Economics /Behavioral economics/allied social sciences; humanities and management in relevant disciplines; or equivalent with a minimum 55% marks in aggregate or equivalent CGPAas determined by the DTU
OR
Bachelor's degree in Engineering/Technology in relevant discipline or equivalent with a minimum 75% marks in aggregate or equivalent CGPA and having proven research capability.</td></tr>
<tr><td>Innovation, Entrepreneurship & Venture Development</td><td>Master's degree in Management/Entrepreneurship/ allied areas related to innovation, venture development and in relevant disciplines, or equivalent, with a minimum 55% marks in aggregate or equivalent CGPA as determined by the DTU
OR
Bachelor's degree in Engineering/Technology in relevant discipline or equivalent with a minimum 75% marks in aggregate or equivalent CGPA and having proven research capability.</td></tr>
<tr><td>Design</td><td>Design</td><td>Master's degree in Design or equivalent with a minimum 55% marks in aggregate or equivalent CGPA as determined by DTU.</td></tr>
</table>

Academic Department	Disciplines Offered	Discipline Specific Eligibility Criteria
Centre of Excellence for the Science of Happiness (CESH)	Science for Happiness	Master's degree in Engineering/Technology/ Science/Management/ Social Science /Arts/Humanities/ Psychology /Medicine and other behavioral Sciences and allied relevant disciplines or equivalent, with minimum 55% marks in aggregate or equivalent CGPA as determined by DTU OR Bachelor's degree in Engineering/Technology or equivalent with a minimum 75% marks in aggregate or equivalent CGPA and having proven research capability.
Department of Geospatial Science and Technology (DGST)	Geoinformatics	Master's Degree in Engineering/ Technology or equivalent in any branch/ discipline with a minimum 55% marks in aggregate or equivalent CGPA as determined by DTU OR Master's degree in Computer Applications/Sciences or equivalent in any branch/discipline with a minimum 55% marks in aggregate or equivalent CGPA as determined by DTU OR Bachelor's degree in Engineering/Technology in relevant discipline or equivalent with a minimum 75% marks in aggregate or equivalent CGPA and having proven research capability
Vinod Dham Centre of Excellence for Semiconductors and Microelectronics (VDSemiX)	Semiconductors and Microelectronics	Master's degree in Engineering / Technology / Sciences in relevant disciplines or equivalent with a minimum 55% marks in aggregate or equivalent CGPA as determined by DTU OR Bachelor's degree in Engineering/Technology in relevant discipline or equivalent with a minimum 75% marks in aggregate or equivalent CGPA and having proven research capability.
Centre for Community Development & Research (CCDR)	Community Development and Research	Master's degree in Engineering/Technology/ Science/Management/Social Science /Arts/Humanities/ Psychology /Medicine and other behavioural Sciences and allied relevant disciplines or equivalent, with minimum 55% marks in aggregate or equivalent CGPA as determined by DTU. OR Bachelor's degree in Engineering/Technology or equivalent with a minimum 75% marks in aggregate or equivalent CGPA and having proven research capability.

In addition to the above Discipline Specific Eligibility Criteria, the following admission guidelines are applicable to all the Disciplines/ Departments:

1. Five percent (5%) relaxation in marks at the level of qualifying degree will be given SC/ST/OBC-NCL/Differently-Abled/ EWS category students for determining the eligibility conditions to apply for Ph.D. admission.
2. Part-time applicants will be considered as per 4.2 and will be eligible if
   - the applicant possesses the minimum eligibility qualifications for the degree as given as per 5.
   - the applicant proves to the satisfaction of the University that his official duties permits him to devote sufficient time to research;
   - facilities for research are available at the applicant's place of work in the chosen field of research or the applicant can spare sufficient time to pursue research in the Department of the Delhi Technological University on daily basis, residing in the vicinity of the University; and
   - he/she will be required to reside at the Delhi Technological University for a period of not less than 6 months during his admission period for the Ph.D. programme. (This condition of minimum residence will be automatically waived for candidates who are working in National Capital Region (NCR) of Delhi

or in organisations / institutions located within a distance of 50 km from the Delhi Technological University).

3. The candidate seeking admission to Ph.D. programme as project fellow at Delhi Technological University may be given administrative clearance to seek admission on full time / part time basis subject to recommendations of the concerned Principal Investigator of the said project/centre and approval of the Vice Chancellor, if the candidate is selected in project as project fellow through proper selection. However, he/she must fulfil the minimum eligibility criteria. No relaxation will be given in entrance test.

4. Permanent academic staff of the Delhi Technological University (Including the academic staff of erstwhile Delhi College of Engineering) may be given administrative clearance to seek registration on part-time basis subject to recommendations of the concerned head of the department and approval of the Vice Chancellor. However, the applicant must fulfil minimum eligibility criteria. No relaxation will be given in entrance test, except for the special cases as given below at 6(3) in selection procedure.

6. SELECTION PROCEDURE

1. The short listing of applications possessing the minimum educational qualifications, for the purpose of entrance test & interview will be done by respective departments in consultation with the Ph.D. Coordinator nominated by the Vice-Chancellor. The entrance test will be of 90 minutes duration comprising of 60 multiple choice questions. It will be given a weightage of 70% in result preparation.

   The mode of examination shall be Computer Based Test at DTU Campus. The entrance test syllabus shall consist of 50% of research methodology, and, 50% shall be the subject specific. This is with reference to the UGC regulations, 2022 issued through Gazette notification dated 7th November, 2022.

   The Departmental Research Committee (DRC) of the respective department shall conduct the Interview and presentation which will be given weightage of 30%. The final selection will be recommended by University level committee after combining the 70% weightage to the Entrance test and 30% weightage to the performance in the interview/viva voce for all department.

2. Entrance Test shall be waived only for UGC-NET/UGC-CSIR-NET/DBT/ICMR/IARI-JRF full time fellowship holders, qualified students in GATE/CEED/NET and similar national level tests conducted by equivalent regulatory bodies, all foreign national students including those sponsored by ICCR/MHRD/ MEA, and covered under MoU with Delhi Technological University, and faculty of the Delhi Technological University/ Industry working professional.

   (ii) Entrance Test shall be waived for the UGC-NET candidates (having valid NET Score released by NTA) for the following three categories:

Category-1	Admission to Ph.D. with JRF and appointment as Assistant Professor
Category-2	Admission to Ph.D. without JRF and appointment as Assistant Professor
Category-3	Admission to Ph.D. programme only and not for the award of JRF or appointment as Assistant Professor

   NET score shall be used for admission of Ph.D. programme. For students who qualify in categories 2 and 3, 70% weightage will be given for test scores and 30% weightage for the interview. The Ph.D. admission will be based on the combined merit of NET marks and the marks obtained in the interview/viva voce.

3. Exemption Entrance Test for DTU Non-Teaching Staff:

   Regular non-teaching staff of DTU with a minimum of three (03) years of continuous service shall be exempted from the Ph.D. Entrance Test. Such candidates shall be permitted to directly appear for the interview, subject to the submission of a No Objection Certificate (NOC) from the competent authority of DTU.

   These candidates shall be registered as part-time Ph.D. Scholars and will not be eligible for DTU fellowships. Additionally, they must fulfil the academic eligibility criteria as prescribed in the DTU Ph.D. Admission Brochure, as revised from time to time.

4. Exemption of Entrance Test for officials of Govt. of India/Govt. of NCT of Delhi:

   Officials (academic/non-teaching) under Government of India or the Govt of NCT of Delhi, with a minimum of five (05) years of regular service shall also be exempted from the Ph.D. Entrance Test. Such candidates shall be allowed to directly appear for the interview, subject to the submission of a No-Objection Certificate (NOC) from the competent authority of their respective organization.

   They shall be registered as part-time Ph.D. Scholars and shall not be eligible for DTU fellowships. Furthermore, they must meet the academic eligibility requirements as outline in the DTU Ph.D. Admission Brochure, as revised from time to time. This exemption shall be subject to the approval of the Hon'ble Vice Chancellor, DTU

5. Students with JRF will also be exempted from the entrance test for admission in Part Time Ph.D. programme without fellowship.

6. Students who have secured 50% marks in the entrance test are eligible to be called for the interview. A relaxation of 5% marks will be allowed in the entrance examination for the candidates belonging to SC/ST/OBC-NCL/differently-abled category and Economically Weaker Section (EWS). However, DTU may decide the number of eligible students to be called for the interview based on the number of Ph.D. seats available.

7. Eligible students and exempted category students from the entrance test shall be considered for admission to the Ph.D. programme with 50% marks as the qualifying cut-off in the interview/presentations. A relaxation of 5% marks will be allowed in the interview/presentations for the candidates belonging to SC/ST/OBC-NCL/differently-abled category and Economically Weaker Section (EWS).

8. The merit list for the unreserved category (UR) seats will comprise all the applicants in order of merit. In other words, the merit list will also include SC / ST / OBC-NCL / EWS applicants, irrespective of their category, if they meet the eligibility criteria of merit for UR category. No applicant can be excluded from the UR category merit list just because the applicant belongs to or has applied under SC/ST/OBCNCL/EWS category.

9. All the shortlisted candidates on the basis of entrance test are required to come along with 7 to 8 slides of power point presentation on the topic of their interest during the interview.

7. RESERVATION OF SEATS

1. Reservation of seats with fellowships for applicants in each of the categories of the research scholars shall be in accordance with the policies of Govt. of NCT of Delhi. The percentage of reservations for various categories and relaxation in minimum eligibility conditions as applicable for the academic session 2026-27 is tabulated below.

S. No.	Category	Seats reserved	Relaxation in Essential Qualification
1	Schduled Castes (SC)	15%	5%
2	Scheduled Tribes (ST)	7.5%	5%
3	OBC-NCL	27%	5%
4	Persons with Disability	5% (Horizontal)	5%
5	EWS	10%	5%

The reservations for persons with disabilities will be implemented department wise. Candidates seeking admission must fulfil the eligibility conditions as detailed earlier. The 5% reservation horizontally in seat matrix for persons with disability may be allocated as follows:

Against the seats identified for each disability, of which, one percent each shall be reserved for persons with benchmark disabilities under clauses (a), (b), and (c) and one percent, under clauses (d) and (e).

a. Blindness and low vision;

b. Deaf and hard of hearing;

c. Locomotor disability including cerebral palsy, leprosy cured, dwarfism, acid attack victims and muscular dystrophy;

d. Autism, intellectual disability, specific learning disability and mental illness;

e. Multiple disabilities from amongst persons under clauses (a) to (d) including deaf-blindness.

2. Physically handicapped applicants are permitted 5% marks or equivalent CGPA relaxation in eligibility requirement in line with the policies of Govt. of NCT of Delhi. They will not be allowed any other relaxation beyond this limit even if they belong to SC/ST category

3. Detailed seat matrix (i.e., seats with DTU fellowship) indicating seats in various departments is available at Annexure-1.

8. APPLICATION PROCESS

For admission to Ph.D. programme August 2026, all candidates need to register and fill the application ONLINE only by accessing www.dtu.ac.in from May 18, 2026 to June 5, 2026. The application process is completed only when a print out of the filled ONLINE application form is taken after paying online the registration fee. The candidate must bring a duly signed copy of the same along with two good quality photo (same as uploaded on online application form) affixed in the appropriate place on the form on the day of interview.

Candidates are requested to ensure that they must fulfil all such requirements before filling and applying for such programmes as their choices. Incomplete application due to any reason is liable for rejection by the University. In this regard, no communication will be entertained.

Application Fee

The registration fee of Rs. 1500/- for all categories is to be paid online through credit/debit card /net banking at the time of registration and choice filling. The registration shall not be complete without the payment of registration fee which is non-refundable and would not be adjusted towards any other fee. A convenience charge (online transaction) will be extra as per banking gateway on every online registration fee payment. If a candidate wishes to apply for admission in a programme offered by different departments then he/she will have to register separately in that department by paying separate online registration fee.

9. Information about Academic Departments

1. Department of Applied Chemistry

The department aims to provide state-of-art knowledge and practical skills to the UG and PG students in the diverse subjects of Chemistry, Chemical Engineering and Polymer Technology. The department runs four year course of B. Tech Chemical Engineering and postgraduate course in Chemistry and Polymer Technology. The Department has well–established laboratories in Applied Chemistry, Polymer Science and Chemical Technology. The department has undertaken and completed successfully large numbers of research and industry projects funded by AICTE, CSIR, UGC, DRDO, DST, BARC etc.

Active national and International collaborations for R&D activities in different fields have been established by the department.

Research Areas: Chemistry including synthetic organic chemistry, Bio inorganic chemistry, Bio organic chemistry, Inorganic Chemistry, Material Chemistry, Green Chemistry Cheminformatics; Medicinal Chemistry; including gene delivery applications, Bio Materials, Drug Delivery systems; Polymer Science including fiber Technology, Conducting Polymers, Composites, Hydrogels: Chemical Engineering including Reaction engineering, Multi phase reactor systems and design, Pollution abatement technology and gene; Advance materials development, Separation processes, Transport Phenomena, Pharmaceuticals sciences, Food Science.

2 Department of Applied Mathematics

The Department runs a four-year B. Tech. programme in Mathematics & Computing. This programme is an amalgamation of Mathematics with Computer Science and Financial Engineering. More than 25 research students are registered in the Department for Ph. D programme. The department has a team of committed faculty members from the disciplines of Pure Mathematics, Applied Mathematics, Computer Engineering, Statistics and Operation Research.

Research Areas: Information Theory, Graph Theory, Discrete Mathematics, Numerical Analysis, General Relativity and Cosmology, Optimization Technique, Complex Analysis, Mathematical Modelling, Approximation Theory, Stochastic Processes, Fuzzy logic and optimization, Algebra and Mathematical Finance, Natural Language Processing (NLP) and Artificial Intelligence (AI).

3 Department of Applied Physics

Applied Physics Department is providing cutting edge research, innovation and education in the emerging areas of science and technology. Department offers the undergraduate academic programme in Engineering Physics and Postgraduate program and M.Sc. Physics. The department has well-equipped state of art laboratories for undergraduate, postgraduate and Ph.D. students. Faculties of the department are actively involved in National and International collaborations for R & D activities.

Research Areas: Nanotechnology: Carbon Nanotube / Carbon Nano fibre and Graphene. Plasma Physics/ Dusty plasma / THz Radiation Emission / High power microwave devices, Photonics and Photonic Crystals. Theoretical Condensed Matter Physics, Density Functional Theory, Heusler alloys based materials for Spintronics and energy application, Topological insulators and Low dimensional Systems. Glass Science and Technology Phosphors, Photoluminescence, Organic & Nano - Material, Time - resolved spectroscopy, Microelectronic Devices-FinFETs, Tunnel FETs, Nanowires, MOSFETs - Application Oriented Modeling and Simulation, Waveguide based devices. Fibre and Integrated optics, Luminescent Material, Material science, Experimental Lithium Ion battery, Multiferroic materials, Atomic physics, Gas sensors, Atmosphere Science, Memory Devices, CNTFET and Graphene FET Devices, CNTFET based Biosensors and Solar energy materials.

4. Department of Biotechnology

The main objective of the department is to provide academic training and conduct research in the interdisciplinary areas of biotechnology with particular emphasis on extending the knowledge generated from these studies towards the development of technologies of commercial significance.

The Department is running postgraduate programmes in Bioinformatics and Industrial Biotechnology. Department of

Biotechnology is also running research oriented Ph.D. programme. The department has undertaken sponsored projects funded by ICMR, CSIR, DST, UGC, etc. The department has 10 state-of-the-art laboratories.

Research Areas: Aquaculture, Algal Biotechnology, Bioremediation, Biosensor, Functional Genomics, Genome informatics, Immunology, Immunostimulation, Molecular Neuroscience, Nano biotechnology, Neuro-oncology, Radiation Biology, Water Quality Management.

5. Department of Civil Engineering

The Department offers an undergraduate programme with an intake of about 120. The post graduate Programmes are offered in Hydraulics and Water Resource and Engineering, Structural Engineering, Environmental Engineering, and Geotechnical Engineering with the intake of about 90 students. The Department is well equipped with laboratories related to Structure, Concrete testing, Soil Mechanics, Highway Engineering, Experimental Stress Analysis, etc. The department lays greater emphasis on quality research of industrial design and development, Structural Engineering and Structural Dynamics.

Research Areas: Structural Engineering, Concrete Technology, Cementitious Materials, Prestressed Concrete, Tall Structures and Rehabilitation of Structures, Geotechnical Engineering, Rock Mechanics, Soil Mechanics, Geo- Environment Engineering, Water Resources Engineering, Pavement Engineering. Hyper Spectral Microwave and LIDAR Remote Sensing.

6. Department of Computer Science and Engineering

The Department of Computer Science and Engineering endeavours to provide the thrill of a corporate and R&D environment with a planned focus on industrially relevant projects and technology incubation. The department has been offering two M.Tech level programs in Computer Science and Engineering and Artificial Intelligence. The department has elite faculties from premier Institutes. Department has developed state of the art laboratories in the various fields of Computer Engineering–Computer Architecture Lab, Network Lab, Web Designing Lab, Computation and Programming Lab, Operating System Lab, Artificial Intelligence Lab, Machine Learning Research Lab, Internet of Things Lab and many others. The department offers Doctoral (Ph.D) degree programs in diverse recent areas and a large number of problems have been taken up in those collaborations (like Samsung Research Lab etc.) with industries. The department has successfully fetched projects for National and International organizations.

Research Areas: Machine Learning, Artificial Intelligence, High Performance Computing, Mobile Computing, Soft Computing, Optimisation techniques, Parallel Computing, Cloud Computing, Internet of Things, Wireless Sensor Networks, Quantum Computing, Block Chain, Nature Inspired Optimisation, Virtual and Augmented Reality, Web Technology, Image Processing, Evolutionary Computing, Big Data, Computer Vision, Steganography, Network Security, Information Security, Software Defined Networks, Software Engineering.

7. Delhi School of Management

DSM carries forward the rich legacy of Delhi College of Engineering by nurturing techno-managers equipped to tackle complex business challenges through analytical thinking and managerial expertise. It offers MBA, MBA (Executive), MBA (Business Analytics), MBA (Innovation, Entrepreneurship & Venture Development), Ph.D., and post-doctoral programs. With a strong emphasis on research, DSM fosters a culture of knowledge creation, outcome-based learning, and industry-oriented education. It is committed to shaping future leaders through experiential learning, interdisciplinary collaboration, and exploration of emerging management trends.

Research Areas: Managerial themes such as E-Governance, Information Technology Management, Marketing Management, Distribution and Retail Management, Brand Management, Organizational Behaviour & Human Resource Management, Corporate Governance and Ethics, Public Policy and Governance, Accounting and Finance (including but not limited to CEO succession, Accounting Theory, Accounting Standards, Directors' Remuneration, Valuation of Human Resources and Intangibles), Portfolio Management, Mergers and Acquisition, Corporate Restructuring, Knowledge Management, International

Business and Trade, Supply Chain Management and Operations Management, Strategic Management, Business Analytics, Entrepreneurship Management.

8. Department of Design

DTU has started Department of Design from academic year 2018-19 with a vision of pursuing excellence in design thinking, design research and design practice for the betterment of society or the ecosystem we live in. The department of design would concentrate on developing a knowledge based design professionals who would become the problem solvers, and who can effectively use different design methodology. They would develop their innovative and aesthetic sensibilities into making a coherent and appropriate research. To develop centres of excellence in design research and practice, the department aspires to initiate Ph.D. programme as per the expertise and strength of the department.

Research Areas: Product Design, Industrial Design, Visual Communication, Interaction Design.

9. Department of Electrical Engineering

The goal of the department is to provide quality education at undergraduate and post graduate levels and undertake cutting edge research in various areas of Electrical Engineering. The department also aims to develop active collaboration with various industries in the power sector. The department has an annual intake of 300 and 60 students in the B.Tech programmes in Electrical Engineering and B.Tech (Evening), respectively. At the post graduate level, the department is offering three M.Tech programmes in Control and Instrumentation, Power Systems and Power Electronics and Systems. In addition to the above the department offers regular Ph.D. programmes in various areas of specialization in Electrical Engineering. The department is involved in carrying out several sponsored R&D projects funded by national agencies like AICTE, DST etc. The department is establishing a new **Centre of Excellence for Electrical Vehicle and Related Technologies (COE for EVRT)** which is funded jointly by Govt. of NCT of Delhi and Delhi Technological University.

Research Areas: Power system optimization, AI Techniques, Modelling & Analysis of Electrical machines, Power Electronics & Drives, Intelligent control of nonlinear systems, FACTS, SSR, Voltage stability, Power quality improvement, Grid integration, Micro grid, Smart grid, Analog Signal processing (Linear and Non linear), Power system & control, System Engineering, Power System Analysis, Power electronics, Renewable energy, HVDC, Power systems restructuring, AI in Electricity market forecasting, Wind energy forecasting, Embedded system, Information security, Design of power supply, Electric traction systems, Energy conversion, IOT enabled electrical system, Charging infrastructure for EVs, Battery management system (BMS), Electric drives & control, EV retrofitting.

10. Department of Electronics and Communication Engineering

The vision of the department is to foster education, innovation and research in the frontline areas of Electronics and Communication Engineering for the sustainable growth of nation and service to the mankind. The department offers UG and PG programmes with annual intake of 240 students in the B. Tech programme in Electronics and Communication Engineering and the PG Programmes include M. Tech. in VLSI Design and Embedded Systems; Signal Processing and Digital Design; and Microwave and Optical Communication. The Department has focused attention on quality research and offers Ph. D. Programmes in the area of Electronics and Communication namely VLSI, DSP, Image Processing, Micro strip antenna design, Sensor Networks, Analog and digital system design. The Department also has active MoUs with academic institutions, research labs and the industrial sector to ensure that the students and faculty can get ample opportunities to work on real-world problems in collaboration through these MoUs.

Research Areas: VLSI Design, Semiconductor Devices Computer Vision, Pattern Recognition, Object Tracking, Image Processing, Machine learning, Artificial Intelligence Human Computer Interaction, Wireless Sensor Network, Microwave Engineering, Antenna Design, Digital Signal Processing, Wireless Communication, RF Devices, Nano-electronics, Network Security, and Cloud Computing, Optical Communication, R F Circuit Design.

11. Department of Environmental Science and Engineering

The Department of Environmental Science and Engineering have been offering M. Tech. in Environmental Engineering Degree and B. Tech in Environmental Engineering. The department conducts cutting-edge research, in developing the vital areas that address societal needs for environmentally sustainable life style and offer doctoral degree. In addition to this, the department also provides opportunity to working engineers for upgrading their qualification under Continuing Education Programme on part time basis for PG level. The departmental laboratories (Water Pollution, Air & Noise Pollution, Microbiology, GIS & Remote Sensing and Solid Waste) for teaching and research are among the best in the nation, providing opportunities for hands- on experience for all students.

Research Areas: Water Pollution, Waste Water Treatment, Environment Modelling, Phytoremediation, Water Management, Air Pollution, Geo- environmental Engineering, Solid Waste Management and Noise Pollution.

12. Department of Humanities

The Department of Humanities offers courses in Communication Skill, English, Economics and Accountancy for engineering and management students in an effort to train them for the global economic environment of the 21st Century. Besides giving them in-depth understanding of the labour market in which they are expected to work as well as of emerging employment trends among engineers, students are sensitized towards the specific technological needs of urban slums and rural areas and socio economic impact of engineering projects on the masses. The Department has competent faculty members with a high degree of excellence who keeps pace with the current developments in their fields of specialization for the fulfilment of its teaching and research goals. More than 20 research students are registered in the Department for Ph.D. programme.

Research Areas:

a) Economics: Women Education and Inclusive Growth, Banking and finance, International Trade and other areas of economics.

b) English: Contemporary Fiction, Cultural Studies, Diaspora Studies, Post-modern literature.

13. Department of Information Technology

The Department of Information Technology (IT) endeavors to provide the thrill of a corporate R&D environment with a planned focus on industrially relevant subjects, projects and technology incubation. The department offers an undergraduate (B. Tech.) course in Information Technology with an intake of 180 students every year. To meet the growing demands of present day technologies, the department has started M.Tech. degree in Information Systems in 2009-10 with an intake of 25. The curriculum of the department is designed in a way so as to provide the students with fundamental concepts and learning tools related to outcome based studies. The designed courses mainly emphasize on all basic core subjects such as data structure, operating systems, computer architecture and design, software development, networking, multimedia and graphics, analog and digital communications and computer communications, compiler design, theory of computation, etc. The department is also imparted a specialized subject knowledge on analysis and design of Information system, Information security systems, mobile computing, Internet of Things, cloud computing & security, soft computing, artificial intelligence, digital signal processing, computer vision and expert systems and web engineering. The department has developed the various state-of-the-art laboratories in the fields of Information Technology such as Computer Networks Lab, Web Engineering Lab, Programming Lab, Information and Security Lab, Biometric Research Lab, and many others.

Research Areas: Pattern Recognition, Computer Vision, Soft computing, Biometric security system, Neural Networks/ Deep learning, Fuzzy-Neural Networks, Natural Language Processing, Optimizations Techniques, Computer Vision, Big data Analytics Web Mining Internet Technologies, Data Mining Social Networks, Social Media Mining, Social Computing, Human behaviour, Multimedia Systems Human Computer Interaction (HCI), Image processing, Human Action and Activity Recognition, Sentiment Analysis, Spam Analysis, Fake News Analysis, Rumour Detection, Evolutionary Computing, Wireless Ad-hoc & Sensor Networks, Internet of Things (IoT), Software Defined Networking (SDN), Network Security, Information Security, Mobile Security, Internet of Robotics Things (IoRT), Cyber Physical System Security, Flying Ad-hoc Network (FANET) Security, Distributed Computing, Pattern Mining and Digital Forensics, Blockchain Technology, Recommendation Systems, Affective Computing, Autonomous Vehicles, etc.

14. Department of Mechanical Engineering

The Department of Mechanical Engineering also offers undergraduate and Postgraduate courses with specialization in

a) Thermal Engineering

b) Production Engineering.

c) Industrial Engineering and Management

d) Computer Aided Analysis and Design

e) Energy Systems and Management

Ph. D Programmes in all fields of Mechanical Engineering are also offered. The department possesses modern laboratories equipped with latest experimental set-ups and research facilities for instrumentation, experimental stress analysis, strength of materials, fluid mechanics, tic, engines, automotive engineering, robotics, heat transfer, solar energy, flexible manufacturing system, computational fluid dynamics supported by Software like view-flex, CAD-CAM and I.e. engine design. Cad lab has Softwares like NX-LAD, NXCAM, AUTOCAD Inventor, Catia, Techomatix, Abus, Iadino, NX-Narran, Hyper mesh, hyper works, MDADAMS, Dynaform etc. The department has many research projects which are sponsored by different government organizations.

Research Areas: Turbo Machinery, Fluid Me chanics, Power Plant b Engineering, Refrigeration and Air conditioning, Computational Fluid Dynamics, Solar Energy, Bio Fuels, Power Plant, Industrial Engineering & Supply Chain Management, Robotics, CAD/CAM, Welding, Production Engineering, System Dynamics, Structural Vibration, Modeling & Simulation, Automation, Advanced Manufacturing Process, Human Factor Engineering, Quality Engineering.

15. Department of Software Engineering

The Department of Software Engineering is dedicated to produce high quality graduates and skilled software engineers/professionals who can develop high quality and cost-effective software systems.

The Discipline of Software Engineering was introduced in the year 2009. The Department is currently running a B.Tech program in Software Engineering with an intake of 180, two M.Tech programs in Software Engineering and Data Science and offers Ph.D. in the Discipline of Computer Science and Software Engineering. All the software engineering programs are well designed keeping in view the industry demands. The programs are designed to build the analytical and practical capabilities of students in the design and development of the software and lays emphasis on following well defined and systematic approach for meeting the growing demands and requirements of the software industry. The department has state of art labs consistent with industrial standards which provide a hands-on experience to the students.

Research Areas: Empirical software engineering, machine learning, software quality and testing, search based software engineering, web engineering, opinion mining, social web, predictive modeling, machine learning and Deep learning for mobile healthcare, telemedicine, internet of things, cryptography.

16. USME

USME (University School of Management and Entrepreneurship) The Vision of the USME is to develop and nurture the spirit of management leadership and entrepreneurship for the good of society. The mission focuses on a portfolio of Programmes around entrepreneurship and cutting edge areas in management, and offers courses at the undergraduate and postgraduate levels: Executive MBA (Data Science and Analytics MA (Economics), BA (Hons.) Economics and BBA (Hons.) four year degree and BBA (IEV).

The focus of the department is to create a practice school in the area of management which is based on research and leads back to Programmes with a strong focus on entrepreneurship, employability, skill development and holistic, experiential learning.

Research Areas: The research focus of USME faculty covers diverse areas within management, economics and analytics. The management research focus is in the areas of work performance management, CRM and behavioural models. The research interests in the area of analytics are in optimization and multi-criterion decision models, quantitative models of innovation diffusion and analytics, social networks and collective intelligence. Faculty in economics stream have research focus in the area of international banking and market structures and health economics and capital markets.

17. Centre of Excellence for the Science of Happiness (CESH)

Established in alignment with NEP 2020, the Centre of Excellence for the Science of Happiness aims to promote holistic development through high-quality research and education in happiness, meditation, and value-based learning. The center offers various Foundation elective and value-added courses and conducts student and staff development programs.

The Centre/DTUhas signed MoUs with several esteemed organizations, including the Rekhi Foundation, Heartfulness Institute, Art of Living, and Darshan Education Academy, to foster collaboration in academic and research initiatives.

The Centre is equipped with a Mind Lab, featuring advanced technologies such as Virtual Reality (VR) headsets, Electroencephalography (EEG) devices, and Embrace Plus smartwatches, aimed at facilitating in-depth research into the cognitive and neuroscientific aspects of happiness.

Research Areas: Workplace happiness, Psychometric tool construction for measuring Happiness, Neuroscience of Positive emotions, Emotional Intelligence, Integration of spirituality in counseling, Ethics and Human values, Positive Psychology, Yoga and Holistic well-being, Meditation and Stress Management, Sustainable development, Indian Psychology

18. Department of Geospatial Science and Technology (DGST)

Geospatial Sciences and Technologies (GST) have emerged as powerful tools to address the challenges of sustainability being faced by the Humanity. By integrating multidisciplinary domains such as geography, geodesy, remote sensing, cartography, GNSS, photogrammetry, geology, environmental sciences, computer science, electronics, and data analytics, GST empowers us to optimize resource management, understand earth systems and processes predict, manage, and mitigate disasters and advance planetary exploration beyond the Earth

Aligned with this National Vision, Delhi Technological University (DTU), with its 82-year legacy of excellence, first established a Multidisciplinary Centre for Geoinformatics (MCG) on 5th March, 2019 with the vision to be a world class multidisciplinary Center in the field of Geospatial education, research and consultancy. The Centre actively contributed to the field of GST. It was due to the efforts of MCG, Geomatics has been added to the GATE Exam wef 2022. The Centre was part of the DST team which worked towards organising the 2nd United Nations World Geospatial Congress (UNWGIC) at Hyderabad, India in the year 2022. It was the experiences gained as a Centre that led the University to further expand the Centre by making it a full-fledged Department wef 2024.

Research Areas : Passive EO (Hyperspectral & Thermal), Active EO – Microwave & SAR, Geodesy, GNSS, Navigation & Communication, Geospatial Data, GIS & AI Analytics, Advanced Drone, LiDAR Surveying & Mapping, Planetary & Solar Mission Data Analytics, Space Science, Astronomy & Astrophysics, Space Systems & Payload Instrumentation, Human Spaceflight & Space Biology.

19. Vinod Dham Centre of Excellence for Semiconductors and Microelectronics (VDSemiX)

The Vision of the Centre is to stimulate and create a robust R & D ecosystem that drives innovation, IP and start-ups in Semiconductor Technology and Microelectronics to cater to the Nation's scientific demands; and serve as a Centre of National and Strategic importance.DTU has received a research donation from its alumnus, Vinod Dham, an Indian-American engineer, entrepreneur, venture capitalist and popularly known as the "Father of Pentium Chip" with a vision to enhance the research capabilities at DTU and foster the development of human resources in the field of semiconductorsin line with the India Semiconductor Mission. VDSemiXwill focus on imparting training & research in thrust areas of Semiconductor Technology and IC Manufacturing; and will provide a platform to boost productivity, address emerging skill gaps and align training& research with industry needs in order to support Government of India's vision to build a vibrant Semiconductor and Display Ecosystem enabling India's emergence as a global hub for electronics manufacturing and design.

VDSemiX has started fostering collaborations and networking by signing Memorandum of Understanding (MoU) with various industries,research laboratories and other institutions of excellence in India and abroad to carry out cutting-edge research and development in Semiconductor Technologies and Microelectronics. VDSemiX has already introduced a Minor specialization in Semiconductors and Microelectronics for B.Tech students from the Academic Session 2023-24, in order to enhance the semiconductor manufacturing skills and increase the spectrum of students pursuing course in Semiconductors so as to build a strong semiconductor and display ecosystem.

This Centre will not only provide interdisciplinary research and development ecosystems but also offer internships, skills development courses, entrepreneurship programs and will enable students develop skills to solve complex technological problems of current times and create world class professionals competent enough in the area of Semiconductors and Microelectronics.

Research Areas : The centre focuses on Semiconductor Device Modeling & Simulation-Analysis of Novel Device Structures-FinFETs, Nanowires, Tunnel FETs, HEMTs for Biomedical, Wireless & Sensor Applications; Novel Material based Devices (III-V Compound Semiconductor, Spintronics, Opto-electronics); Memory Devices; Quantum and AI assisted novel materials; Development of Nanoparticles (Quantum Dots) and 2D structures for QLEDs, Solar Cell and Sensors; Analog and Digital VLSI Design; System Design using FPGAs.

20. Centre for Community Development & Research (CCDR)

The Centre for Community Development & Research (CCDR) at Delhi Technological University represents a pioneering initiative that merges academic rigor with practical community engagement. Our mission is to facilitate sustainable community development through participatory research methodologies, innovative training programs, and strategic collaborations. The following are some of the research areas of CCDR:

1. Community-Based Education Models. Research on service-learning, experiential pedagogy, and integrating community issues into curricula.
2. Youth Development & Student Engagement. Studies on leadership training, entrepreneurship, and volunteerism among university students in local communities.
3. Inclusive Growth & Social Innovation. University-led incubators for social enterprises, microfinance, and skill-building programs.
4. Public Health & Preventive Care. Collaborative research with medical faculties on nutrition, mental health, and community health systems.
5. Sustainable Campus & Community Practices. Linking zero-waste initiatives, renewable energy, and green infrastructure between campus and surrounding communities.
6. Climate Change Adaptation & Resilience. Research on disaster preparedness, sustainable agriculture, and community resilience strategies.
7. Participatory Governance & Policy Labs. University-hosted policy experiments, student-led governance models, and community advisory councils.
8. Migration, Urbanization & Housing Studies. Academic research on rural-to-urban transitions, slum development, and affordable housing policies.
9. Sustainable energy, air and water pollution mitigation and solutions
10. Ethical integration of AI based societal development.

10. Fee and Documents to be submitted

The selected candidates will be required to pay the admission fee as per the details given below:

S. No.	Particulars	Annual Fee			
		At the time of Admission		2nd year onwards	
	Code	A	B	A	B
1	Tuition Fee	Nil	14000	Nil	7000
2	Non Govt. Component				
2.1	Student Welfare Fee (Co-curricular activities, Training & placement, Extra Curricular Activities, Annual Gathering, Students welfare, Institutional Development, outsourcing, conference, seminar, workshop, innovative projects, skill development activities and, Misc. Expenditure on Unspecified Items)	7000	7000	5500	5500
2.2	Facilities & Services Charges (Research initiatives, training programmes, Awards, automation, facilities, entrepreneurship activities and any misc. expenditure on unspecified items)	3000	3000	1500	1500
2.3	Economically weaker section fund	1000	1000	1000	1000
2.4	Examination fee (Examination Infrastructure strengthening, expenditure on examination activities, confidential printing etc.	7000	7000	-	-
2.5	Premium amount for mediclaim of student (per-annum)	700	700	700	700
2.6	Alumni Association life membership, registration fee (one time, non-refundable)	1000	1000	-	-
	GRAND TOTAL	**19700**	**33700**	**8700**	**15700**

S. No.	Particulars	Code
1.	The teaching/non-teaching/academic staff of the Delhi Technological University (including the teaching/non-teaching/ academic staff of erstwhile Delhi College of Engineering) and officers of Department of Technical and Higher Education, Govt. of NCT of Delhi as in R.19 (i)	A
2.	Project staff pursuing Ph.D. as in R.19(ii)	
3.	Other Full Time / Part Time candidates	B

11. Withdrawal / Refund Policy

S. No.	Percentage of Refund of aggregate fee*	Point of Time when Application for Withdrawal of admission is received
1	100%	Application for withdrawal of admission received up to 24.07.2026.
2	90%	Application for withdrawal of admission received from 25.07.2026 to 31.07.2026.
3	80%	Application for withdrawal of admission received from 1.08.2026 to 07.08.2026.
4	50%	Application for withdrawal of admission received from 08.08.2026 to 14.08.2026.
5	NIL	Application for withdrawal of admission received after 15.08.2026.

Original Documents to be submitted for verification at the time of Admission

a. Printed copy of online registration application

b. All the semesters Mark sheet/grade card / provisional / degree certificates beginning from first degree towards proof of qualification.

c. All the candidates will be required to produce the proof of having passed the qualifying degree with the required percentage of marks or CGPA latest by July 31, 2026, failing which their admission shall be cancelled.

d. Caste Certificate in the case of SC/ST/OBC-NCL candidates issued by respective State Government as per format annexed at 2. OBC (NCL) Candidates are required to produces Caste Certificate issued after 31st March, 2026. However, if the certificate is issued prior to March 31, 2026, it must be accompanied with an additional certificate regarding the present non-creamy layer status of the candidate, issued by the same competent authority. This additional certificate must have reference of his/her already issued original caste certificate.

e. Authorised Doctor's Certificate with disability descriptions in the case of Person with Disabled (PwD) candidates as per format annexed at 3.

f. EWS Income and Asset Certificate issued by compliant authority as per Annexure-4 issued after 31st March, 2026.

g. NOC for part time candidates as per format annexed at 5.

h. Original UGC- JRF/NET/CSIR/ DAE-JEST or other fellowship award letter.

i. Project Co-ordinator's certificate in the prescribed format

j. Date of Birth Proof

k. Two passport sized recent photographs

l. Copy of Cancelled Cheque

m. Copy of Adhaar Card

IMPORTANT INSTRUCTIONS

1. The candidates are advised to read each and every instruction given in this Information Brochure very carefully before applying Online.
2. All entries should be carefully made while applying online. DTU will not be responsible for wrong entries. Candidates shall be sole responsible for the correctness and authenticity of the information / documents provided in the online application.
3. Online application found incomplete in any form will be summarily rejected. No correspondence / communication will be entertained in this regard.
4. The last date for submission of online application shall not be extended. Accordingly, no request shall be entertained for accepting the application after the last date. Therefore, candidates are advised to submit their application well in advance and not to wait for the last moment.
5. The University has the right to cancel, at any stage, the admission for the candidate who is found admitted to a course to which he/she is not entitled, being unqualified or ineligible in accordance with the statues and regulations in force.
6. The list of the shortlisted candidates for entrance test, interview and finally selected for admissions to Ph.D. programme will be displayed on the DTU website: www.dtu.ac.in
7. Candidates have to bring a valid photo-identity card for the purpose of written test along with the printed application form.
8. There is no need to send any part of application form to DTU by post.
9. Incomplete applications are likely to be rejected.
10. No separate call letter will be dispatched.
11. The candidates are advised to make their own arrangements for travelling and lodging accordingly. They must come prepared for admission (in case they are selected) as per the schedule.
12. Candidate should check the University website for results / important announcements.
13. Candidates called for the interview should bring with them (i) Photo ID Card, (ii) Printed copy of the application submitted online, (iii) Thesis/ dissertation / report / publications (iv) copy of certificates and mark-sheets.
14. The candidates may contact faculty members/Head of the concerned academic departments for selecting their area of research work.

ANNEXURE-1

Vacant Seats for full time Ph.D. programme with University Scholarship for the Session: January 2026				
S. No.	**Name of the Department**	**Discipline Offered by the Departments**	**Code of the Department**	**Tentative total available Seats with DTU Fellowship**
1	Applied Chemistry	1. Chemistry	AC	3
		2. Chemical Engineering		2
2	Applied Physics	1. Physics	AP	7
		2. Engineering Physics		1
3	Applied Mathematics	1. Mathematics	AM	9
		2. Mathematics and Computing		4
4	Biotechnology	1. Biotechnology	BT	5
5	Design	1. Design	DS	3
6	Delhi School of Management	1. Management	DSM	4
7	Civil Engineering	1. Civil Engineering	CE	9
8	Computer Science and Engineering	1. Computer Science and Engineering	CSE	37
9	Electronics & Communication Engineering	1. Electronics and Communication Engineering	ECE	21
10	Electrical Engineering	1. Electrical Engineering	EE	10
11	Environmental Science and Engineering	1. Environmental Science and Engineering	ENE	5
12	Mechanical Engineering	1. Mechanical Engineering	ME	10
13	Information Technology	1. Information Technology	IT	15
14	Humanities	1. Economics	HUM	0
		2. English		1
15	Software Engineering	1. Software Engineering	SWE	1
		2. Computer Science		1
16	USME	1. Management	USME	4
		2. Economics		4
		3. Innovation Entrepreneurship & Venture Development		2
17	Department of Geospatial Science and Technology (DGST)	Geoinformatics	DGST	2
18	CESH	Science of Happiness	CESH	2
19	VDSemiX	Semiconductor and Microelectronics	VDSemiX	2
20	COE-EVRT	Electrical Vehicle & Related Technologies		2
21	Centre for Community Development & Research	Community Development & Research	CCDR	2
22	Nodal Centre of Excellence in Energy Transition	Energy Science and Engineering	NCEET	2
	Total			**170**

* Non availability of suitable candidate in the discipline of the Department(s), the seat may be converted in other discipline of the Department.

Note:

1. Reservation Policy: OBC-27%, SC- 15%, ST-7.5%, EWS-10%, PWD: 5% Horizontal
2. This is tentative seat allocation. It may vary as per availability of seats in various departments. University reserves the right to allot a supervisor to the selected candidate, in respective Departments.
3. For UGC/CSIR-NET-JRF/Part Time/Without DTU Fellowship, seats are available subject to the availability or slots with prospective supervisors.

ANNEXURE-2

AUTHORITIES WHO CAN ISSUE CASTE/TRIBE CERTIFICATE

SC/ST/OBC candidates should submit certificate issued by any of the following authorities:

District Magistrate/Additional District Magistrate/Collector/Deputy Commissioner/Additional Deputy Commissioner/Deputy Collector/1st Class Stipendiary Magistrate/City Magistrate/Sub-Divisional Magistrate/Taluka Magistrate/Executive Magistrate/Extra Assistant Commissioner/ Chief Presidency Magistrate/Additional Chief Presidency Magistrate/Presidency Magistrate/ Revenue Officer not below the rank of Tehsildar/Sub-Divisional Officer of the area where the candidate and/or his/her family normally resides/Administrator/Secretary to Administrator/ Development Officer (Lakshadweep Island).

(Certificate issued by any other authority will not be accepted.)

Government of................

(Name & Address of the authority issuing the certificate)

Prescribed Format for OBC Certificate

FORM OF CERTIFICATE TO BE PRODUCED BY OTHER BACKWARD CLASSES

This is to certify that Shri / Smt. / Kum. ________________________________

__________ Son/Daughter of Shri / Smt. ________________________________

____________________ of Village/Town ________________________________

District/Division _____________ in the State belongs to the ____________________

_________________________Community which is recognized as a backward class under:

i. Resolution No. 12011/68/93-BCC(C) dated 10/09/93 published in the Gazette of India Extraordinary Part I Section I No. 186 dated 13/09/93.

ii. Resolution No. 12011/9/94-BCC dated 19/10/94 published in the Gazette of India Extraordinary Part I Section I No. 163 dated 20/10/94.

iii. Resolution No. 12011/7/95-BCC dated 24/05/95 published in the Gazette of India Extraordinary Part I Section I No. 88 dated25/05/95.

iv. Resolution No. 12011/96/94-BCC dated 9/03/96.

v. Resolution No. 12011/44/96-BCC dated 6/12/96 published in the Gazette of India Extraordinary Part I Section I No. 210 dated 11/12/96.

vi. Resolution No. 12011/13/97-BCC dated 03/12/97.

vii. Resolution No. 12011/99/94-BCC dated 11/12/97.

viii. Resolution No. 12011/68/98-BCC dated 27/10/99.

ix. Resolution No. 12011/88/98-BCC dated 6/12/99 published in the Gazette of India Extraordinary Part I Section I No. 270 dated 06/12/99.

x. Resolution No. 12011/36/99-BCC dated 04/04/2000 published in the Gazette of India Extraordinary Part I Section I No. 71 dated04/04/2000.

xi. Resolution No. 12011/44/99-BCC dated 21/09/2000 published in the Gazette of India Extraordinary Part I Section I No. 210dated21/09/2000.

xii. Resolution No. 12015/9/2000-BCC dated 06/09/2001.

xiii. Resolution No. 12011/1/2001-BCC dated 19/06/2003.

xiv. Resolution No. 12011/4/2002-BCC dated 13/01/2004.

xv. Resolution No. 12011/9/2004-BCC dated 16/01/2006 published in the Gazette of India Extraordinary Part I Section I No. 210 dated 16/01/2006.

Shri / Smt. / Kum. ____

and/or his family ordinarily reside(s) in the ______________________ District /

Division of ____________________ State. This is also to certify that he/she does not

belong to the persons/sections (Creamy Layer) mentioned in Column 3 of the Schedule to the Government of India, Department of Personnel & Training O.M. No. 36012/22/93-Estt.(SCT) dated 08/09/93 which is modified vide OM No. 36033/3/2004 Estt.(Res.) dated 09/03/2004.

Dated: ______________

District Magistrate / Deputy Commissioner / Competent Authority

Seal

NOTE:

a. The term 'Ordinarily' used here will have the same meaning as in Section 20 of the Representation of the People Act, 1950.

b. The authorities competent to issue Caste Certificates are indicated below:

   i. District Magistrate / Additional Magistrate / Collector / Deputy Commissioner / Additional Deputy Commissioner / Deputy Collector / Ist Class Stipendiary Magistrate / Sub-Divisional magistrate / Taluka Magistrate / Executive Magistrate / Extra Assistant Commissioner (not below the rank of Ist Class Stipendiary Magistrate).

   ii. Chief Presidency Magistrate / Additional Chief Presidency Magistrate / Presidency Magistrate.

   iii. Revenue Officer not below the rank of Tehsildar' and

   iv. Sub-Divisional Officer of the area where the candidate and / or his family resides.

Declaration/Undertaking - for OBC Candidates only

I, ______________________________ son/daughter of Shri ____________________

____________ resident of village/town/city ________________________________

district ______________________ State ______________ hereby declare that I

belong to the ___________________ community which is recognized as a backward class

by the Government of India for the purpose of reservation in services as per orders contained in Department of Personnel and Training Office Memorandum No.36012/22/93- Estt. (SCT), dated 8/9/1993. It is also declared that I do not belong to persons/sections (Creamy Layer) mentioned in Column 3 of the Schedule to the above referred Office Memorandum, dated 8/9/1993, which is modified vide Department of Personnel and Training Office Memorandum No.36033/3/2004 Estt.(Res.) dated 9/3/2004.

Signature of the Candidate

Place: ________________

Date: ________________

Contact No.: ______________

Email ID ________________________

ANNEXURE-3

Person with Disability Sub-Category

For admission to seat reserved for Differently Abled Person (PwD) sub-category, the candidate must produce the following certificates in original at the time of document verification for PwD candidates:

a. A certificate of physical disability issued by a duly notified Medical Board of a District/ Government Hospital set up for examining the physically challenged candidates under the provision of the Person with Disability (equal opportunities, protection of rights and full participation) Act 1995.The certificate should indicate the extent (i.e. percentage) of the physical handicap and should bear the Photograph of the candidate concerned. The certificate should be countersigned by one of the Doctors constituting the Board issuing the certificates.

b. A certificate duly recommended by Vocational Rehabilitation Centre for the Handicapped, 9 - 11 Vikas Marg, Karkardooma, Delhi 110092.

Certificate for Person with Disability

To be issued by Medical Board from Government Hospital

Name of the candidate: Mr./Ms.*____

Father's Name:____

Permanent Address:____

Percentage loss of earning capacity (in words): ____________________________

____

Whether the candidate is otherwise able to carry on the studies and perform the duties

of an engineer/architect satisfactorily: ______________________.

Name of the disease-causing handicap: ______________________________

Whether handicap is temporary or permanent: ___________________________

Whether handicap is progressive or non-progressive: _______________________

The candidate is FIT / UNFIT to pursue further studies.

(*Strike out whichever is not applicable)

Member Member Principal Medical Officer

(Orthopaedic Specialist)

Seal of Office:

Date: ____________________

NOTE:

1. The medical board must have one orthopaedic specialist as its member.
2. Candidate having temporary or progressive handicap will not be considered against the seats.

ANNEXURE-4

Government of.................

(Name & Address of the authority issuing the certificate)

INCOME & ASSET CERTIFICATE TO BE PRODUCED BY ECONOMICALLY WEAKER SECTIONS

Certificate No. __________

Date: __________________

VALID FOR THE YEAR__________

This is to certify that Shri/Smt./Kumari________________________ son/daughter/wife of ______________________________ permanent resident of__________________________ Village/Street________________ Post Office________________ District_________________ in the State/Union Territory_____________________________ Pin Code_______________ whose photograph is attested below belongs to Economically Weaker Sections, since the gross annual income* of his/her family** is below Rs. 8 lakh (Rupees Eight Lakh only) for the financial year___________. His/her family does not own or possess any of the following assets***:

i. 5 acres of agricultural land and above;

ii. Residential flat of 1000 sq. ft. and above;

iii. Residential plot of 100 sq. yards and above in notified municipalities;

iv. Residential plot of 200 sq. yards and above in. areas other than the notified municipalities.

Shri/Smt./Kumari_______________________ belongs to the___________________________ caste which is not recognized as a Scheduled Caste, Scheduled Tribe and Other Backward Classes (Central List)

Name ____________________________ Signature with seal of Office

Designation_______________________

*Note 1: Income covered all sources i.e. salary, agriculture, business, profession, etc.

**Note 2: The term 'Family" for this purpose include the person, who seeks benefit of reservation, his/her parents and siblings below the age of 18 years as also his/her spouse and children below the age of 18 years.

***Note 3: The property held by a "Family' in different locations or different places/cities have been clubbed while applying the land or property holding test to determine EWS status.

INCOME AND ASSET CERTIFICATE ISSUING AUTHORITY

The Income and Asset Certificate issued 'by any one of the following authorities in the prescribed format as given above shall only be accepted as proof of candidate's claim as 'belonging to EWS: -

i. District Magistrate/Additional District Magistrate/ Collector/ Deputy Commissioner/Additional Deputy Commissioner/ 1st Class Stipendiary Magistrate/ Sub-Divisional Magistrate/ Taluka Magistrate/ Executive Magistrate/ Extra Assistant Commissioner,

ii. Chief Presidency Magistrate/Additional Chief Presidency Magistrate/ Presidency Magistrate,

iii. Revenue Officer not below the rank of Tehsildar and

iv. Sub-Divisional Officer or the area where the candidate and/or his family normally resides.

ANNEXURE-5

NO OBJECTION CERTIFICATE

(Required from candidates seeking admission in Ph.D. on Part-Time Basis)
(On the Organization Letter Head)

The undersigned is pleased to permit Mr./Ms.____ who is working in this organization for the last_______years and is presently holding the rank/position of__________________________ for pursuing the Ph.D. programme at Delhi Technological University, Delhi with specialization in the following areas:

1.______________________________________ 2.____________________________________

To the best of my knowledge and belief Mr./Ms.____________bears a good moral character.

If selected for admission, the candidate will be permitted and be granted leave of absence to be present at the Delhi Technological University as required by the academic schedule to attend classes/ research work. He/She will continue to remain in service of this organization for the entire duration of the course.

This organization has standing commitment to the exemplary standards namely ISO/CMM or similar standard of respective area (required for candidates from industry).

Signature of Head of the
Institution/Organization with seal

Place:_____________________________

Date:______________________________

Name______________________________________

Designation ________________________________

Email ID: ____________________________________

Research Methodology (Common for All)

UNIT 1 - Introduction to Research : The concept of research, characteristics of good research, Application of Research, Meaning and sources of Research problem, characteristics of good Research problem, Research process, outcomes, application of Research, Meaning and types of Research hypothesis, Importance of Review of Literature, Organizing the Review of Literature. Types of Research : Types of research, pure (basic, fundamental) and applied research, qualitative and quantitative.

UNIT 2 - Research Design : Meaning, need, types of research design – Exploratory, Descriptive, Casual research Design, Components of research design, and Features of good Research design. Experiments, surveys and case study Research design. Sampling, Data Collection and analysis: Types and sources of data–Primary and secondary, Methods of collecting data, Concept of sampling and sampling methods – sampling frame, sample, characteristics of good sample, simple random sampling, purposive sampling, convenience sampling, snowball sampling, classification and tabulation of data, graphical representation of data, graphs and charts – Histograms, frequency polygon and frequency curves, bell shaped curve and its properties.

UNIT 3 - Statistical Methods for Data Analysis: Applications of Statistics in Research, measures of central tendency and dispersion, Descriptive statistics, Probability functions, Correlation &regression analysis, Statistical inference.

UNIT 4 - Research Report : Research report and its structure, journal articles – Components of journal article. Explanation of various components. Structure of an abstract and keywords. Thesis and dissertations: components of thesis and dissertations. Referencing styles and bibliography. Ethics in Research - Plagiarism -Definition, different forms, consequences, unintentional plagiarism, copyright infringement, collaborative work. Qualities of good Researcher.

UNIT 5 - ICT Tools for Research : Role of computers in research, maintenance of data using software such as Mendeley, Endnote, Tabulation and graphical presentation of research data and software tools. Web search: Introduction to Internet, use of Internet and WWW, using search engines and advanced search tools.

1. Department of Applied Chemistry

Discipline: Chemistry

Chemical periodicity, structure and bonding, concepts of acids, bases and non-aqueous solvents, main, transition and inner-transition group elements and their compounds including properties, organ metallic compounds, cages and metal clusters, bio inorganic chemistry.

Basic principles of quantum mechanics, chemical thermodynamics, electro chemistry, chemical kinetics, photochemical reactions, colloids and surfaces, solid state and polymer chemistry.

IUPAC nomenclature of organic molecules, configuration, conformational isomerism in acyclic and cyclic compounds. Organic reaction mechanism, determination of reaction pathways, common named reactions and rearrangements and applications in organic synthesis. Separation, electro- and thermo analytical methods. Molecular spectroscopy and characterization of chemical compounds by IR, Raman, NMR, EPR, UV-vis, MS, electron spectroscopy and microscopic techniques.

Discipline: Chemical Engineering

Material and energy balance, Chemical Engineering Thermodynamics, Basic concepts of fluid mechanics, flow through pipe, pressure drop calculations, Transport Phenomena, Heat transfer by conduction, convection, radiation, Concepts and design of heat exchangers and evaporators, Basic concepts of mass transfer, Different mass transfer processes and unit operations, Mechanical operations, Chemical reaction engineering: Design of homogeneous and heterogeneous reactors, Chemical process technology, Petroleum refining and petrochemicals. Instrumentation control and optimization, Principles of process economics and cost estimation.

Polymer chemistry, Polymer properties and testing: Determination of molecular weight, thermal, morphological, structural, Mechanical, optical, electrical and environmental properties of polymers, properties and applications of commodity, engineering and specialty polymers, Natural and synthetic rubbers, Polymer blends and composites, bio polymers, Polymer processing: Compression molding, injection molding, blow molding, extrusion, rotational molding, thermoforming, rubber processing.

2. Department of Applied Physics

Discipline: Physics

Section 1: Mathematical Physics

Vector calculus; matrices; similarity transformations, diagonalization, eigenvalues and eigenvectors; linear differential equations: second order linear differential equations and solutions involving special functions; Partial differential equations, complex analysis: Cauchy-Riemann conditions, Cauchy's theorem, singularities, residue theorem and applications; Laplace trans form, Fourier series & analysis; Tensors; Numerical methods.

Section 2: Classical Mechanics

Lagrangian formulation: D'Alembert's principle, Euler-Lagrange equation, Hamilton's principle, symmetry and conservation laws; central force motion: Kepler problem and Rutherford scattering; Periodic motion: small oscillations; coupled oscillations and normal modes; rigid body dynamics; Liouville's theorem; canonical transformations, Poisson brackets, Hamilton-Jacobi equation. Special theory of relativity: Lorentz transformations, relativistic kinematics, mass-energy equivalence.

Section 3: Electromagnetism

Electrostatics and magneto statics, boundary value problems, multipole expansion, Fields in conducting, dielectric, diamagnetic and paramagnetic media, Faraday's law and time varying fields; displacement current; Maxwell's equations; energy and momentum of electromagnetic fields; Propagation of plane electromagnetic waves, reflection, refraction; Electromagnetic waves in dispersive and conducting media.

Section 4: Quantum Mechanics

Wave-particle duality, uncertainty principle; Schrodinger equation; linear vectors and operators in Hilbert space; one dimensional potentials: step potential, finite rectangular well, tunnelling from a potential barrier, particle in a box, harmonic oscillator; two and three dimensional systems: concept of degeneracy; hydrogen atom; Stern-Gerlach experiment, angular momentum and spin; addition of angular momenta; variational method and WKB approximation, time independent perturbation theory; elementary scattering theory, Born approximation; symmetries in quantum mechanical systems; Identical particles; Pauli exclusion principle.

Section 5: Thermodynamics and Statistical Physics

Laws of thermodynamics and their consequences; Macro states and microstates; phase space; ensembles; partition function, free energy, calculation of thermodynamic quantities; classical and quantum statistics; degenerate Fermi gas; black body radiation and Planck's distribution law; Bose-Einstein condensation; first and second order phase transitions, phase equilibria, critical point.

Section 6: Atomic and Molecular Physics

Quantum states of an electron in an atom; Spectra of one-and many-electron atoms; spin-orbit interaction: LS and jj couplings; fine and hyperfine structures; Zeeman and Stark effects; Electronic, rotational and vibrational spectra of diatomic molecules; selection rules; electronic transitions in diatomic molecules, Franck-Condon principle; Raman effect; EPR, NMR, ESR, X-ray spectra; lasers: Einstein coefficients, population inversion, two, three and four level systems.

Section 7: Solid State Physics

Bravais lattices; crystal structures, Miller indices, diffraction methods for structure determination; Reciprocal lattice, bonding in solids; Defects and dislocations; lattice vibrations and thermal properties of solids; free electron theory; band theory of solids; Optical properties of solids; dielectric properties of solid; dielectric function, polarizability, ferroelectricity; magnetic properties of solids; domains and magnetic anisotropy; superconductivity: Type-Iand Type II superconductors, Meissner effect, London equation, BCS Theory, flux quantization, Josephson junctions.

Section 8: Electronics

Semiconductors, electron and hole statistics in intrinsic and extrinsic semiconductors; Hall effect, metal-semiconductor junctions; Ohmic and rectifying contacts; PN diodes, bipolar junction transistors, field effect devices; negative and positive feedback circuits; oscillators, operational amplifiers, Opto-electronic devices, Microprocessor and micro controller basics.

Section 9: Nuclear and Particle Physics

Basic nuclear properties; Nuclear radii and charge distributions, nuclear binding energy, electric and magnetic moments; semi-empirical mass formula; nuclear models; liquid drop model, nuclear shell model; nuclear force and two nucleon problem; alpha decay, beta-decay, electromagnetic transitions in nuclei; Rutherford scattering, nuclear reactions, conservation laws; fission and fusion; particle accelerators and detectors; elementary particles, Quark model.

Section 10: Optics and Fiber Optics

Interference, diffraction, polarization, Guided wave Optics, Guided Wave Structures, Ray analysis, Modes of planar waveguide, Mode theory for optical fibers, Propagation characteristics of step index fibers, graded index fibers, Signal degradation in optical fiber due to dispersion and attenuation, Optical Sources and detectors for optical fiber communication.

Discipline: Engineering Physics

Section 1: Engineering Mathematics

Linear Algebra: Vector space, basis, linear dependence and independence, matrix algebra, eigenvalues and eigenvectors, rank, solution of linear equations- existence and uniqueness.

Calculus: Mean value theorems, theorems of integral calculus, evaluation of definite and improper integrals, partial derivatives, maxima and minima, multiple integrals, line, surface and volume integrals, Taylor series.

Differential Equations: First order equations (linear and nonlinear), higher order linear differential equations, Cauchy's and Euler's equations, methods of solutions in variation of parameters, complementary function and particular integral, partial differential equations, variable separable method, initial and boundary value problems.

Vector Analysis: Vectors in plane and space, vector operations, gradient, divergence and curl, Gauss's, Green's and Stokes' theorems.

Complex Variables: Analytic functions, Cauchy's integral theorem, Cauchy's integral formula, sequences, series, convergence tests, Taylor and Laurent series, residue theorem.

Section 2: Computational Physics

Roots of Non-linear Equation: Roots of equations, Bisection method, Regula Falsi Method or Method of False position, Secant, Newton-Raphson method, Convergence of these methods.

Interpolation: Finite differences and difference operators, Interpolation with equally spaced data points: Newton's forward and backward formulae for interpolation, Central difference: Gauss forward, Gauss Backward, Stirling, Bessels, Everett's formula for interpolation, Interpolation with unequally data points: Divided differences and their property, Newton Divided differences formula.

Integration: Newton-cotes integration formulae, trapezoidal method, Simpson's 1/3-rule,Simpson's3/8-rule,Boole'sandWeddle's Rule.

Numerical solution of ordinary differential equations: Taylor's series method, Picard's method of successive approximation methods, Euler's method, modified Euler's method, Runge-Kutta method, solution of second order and simultaneous differential equations, Application of optimization and variational methods to problem of interest in applied physics.

Section 3: Quantum Mechanics

Wave-particle duality, uncertainty principle; Schrodinger equation; linear vectors and operators in Hilbert space; one dimensional potentials: step potential, finite rectangular well, tunnelling from a potential barrier, particle in a box, harmonic oscillator; two and three dimensional systems: concept of degeneracy Quantum states of an electron in an atom; Spectra of one-and many-electron atoms; spin-orbit interactions; fine and hyperfine structures; Zeeman and Stark effects; Electronic, rotational and vibrational spectra of diatomic molecules.

Section 4: Engineering Materials

Classification of Materials, Nature of bonding in Materials, Defects in Crystalline Materials, Mechanical, Structural, Electronic, Thermal, Optical Properties and various Applications of Materials.

Section 5: Synthesis and Characterization of Materials

Top-down and bottom-up synthesis approach, physical and chemical techniques for material synthesis (sol-gel, hydrothermal etc.), X-Ray Diffraction, Scanning Electron Microscopy, Transmission Electron Microscopy, Atomic Force Microscopy, Photoluminescence Spectroscopy and other spectroscopic techniques.

Section 6: Electronic Devices and Circuits

Semiconductors, Metal-semiconductor junctions; Ohmic and rectifying contacts; PN diodes, bipolar junction transistors, field effect devices; negative and positive feedback circuits; oscillators, operational amplifiers, active filters; basics of digital logic circuits, combinational and sequential circuits, flip-flops, timers, counters, registers, A/D and D/A conversion, Opto-electronic devices, Microprocessor and microcontroller basics.

Section 7: Communication systems

Random processes: autocorrelation and power spectral density, properties of white noise, filtering of random signals through LTI systems.

Analog communications: amplitude modulation and demodulation, angle modulation and demodulation, spectra of AM and FM, super heterodyne receivers.

Information theory: entropy, mutual information and channel capacity theorem.

Digital communications: PCM, DPCM, digital modulation schemes (ASK, PSK, FSK, QAM), bandwidth, inter-symbol interference, MAP, ML detection, matched filter receiver, SNR and BER.

Section 8: Electromagnetics

Maxwell's equations: Differential and integral forms and their interpretation, boundary conditions, wave equation, Poynting vector.

Plane waves and properties: reflection and refraction, polarization, Phase and group velocity, propagation through various media, skin depth.

Transmission lines, Rectangular and circular waveguides, light propagation in optical fibers, dipole and monopole antennas, linear antenna arrays.

3. Department of Applied Mathematics

DISCIPLINE: MATHEMATICS

Differential & Integral calculus Vector calculus Algebra (Linear & Abstract) Differential equations (ODE & PDE) Laplace &Fourier transforms, Fourier series. Probability, statistics & operations research Numerical methods, Special functions Real and complex analysis.

DISCIPLINE: MATHEMATICS AND COMPUTING:

Entrance exam will be 30% in Mathematics domain and 70% in Computing Domain

Mathematics Doman (30%)

Calculus: Sequences and Series, differential calculus, integral calculus, vector calculus. Basics and ordinary differential equation (ODE) and partial differential equation (PDE), Fourier Series.

Linear Algebra: Determinants and matrices, Cayley-Hamilton Theorem, Hermitian, skew Hermitian, unitary matrices, eigen values, eigen vectors.

Transforms: Laplace Transform, Fourier Transform.

Numerical Analysis: Numerical solution of algebraic equations using Gauss elimination and Gauss-Siedel methods, Gauss Jordan, numerical solution of ordinary differential equations using Picard, Euler method. Interpolation.

Probability and Statistics: mean, mode, median and standard deviation, Probability space, conditional probability, Baye's theorem, uniform, binomial, poisson, normal and exponential distribution.

Computing Domain (70%)

Programming: Programming fundamentals & C/C++ programming, object-oriented programming concepts using C++/JAVA.

Data Structures and Algorithms: Arrays, stack, queue, linked list, trees, binary search trees, graph, sorting and searching. Algorithm design techniques: greedy, dynamic programming and divide-and-conquer. Asymptotic space and time complexity.

Digital Logic Design: Boolean Algebra, Logic Functions, combinational and sequential circuit design, registers and counters, logic families.

Computer organization and architecture: Machine instructions and addressing modes, Instruction pipelining, memory hierarchy, I/O devices.

Computer network: OSI, TCP/IP, Brief overview of OSI layers, IPv4, IPv6, routers and routing algorithms (distance vector, link state), congestion control. Application layer protocols (DNS, SMTP, POP, FTP,HTTP).

Database Management System: ER model, relational model, normalization, integrity constraints, SQL, indexing transaction processing, concurrency control.

Theory of Computation: Regular expression, Finite automata, context free grammar, pushdown automata, regular and context free languages, pumping lemma, Turing machine.

Compiler Design: Lexical Analysis, parsing, syntax-directed translation, runtime environments, intermediate code generation.

Operating System: Types of OS, process management, concurrency and synchronization, CPU scheduling, deadlocks, memory management and virtual memory, disk scheduling, file systems.

Discrete Structures: Classical logic theory, Boolean algebra and relations and graph, tree, spanning tree, planar graph and coloring.

Software Engineering: Requirement and Feasibility Analysis, Data flow Diagrams, Process life cycle, design, coding, testing, implementation and maintenance.

4. Department of Civil Engineering

DISCIPLINE: CIVIL ENGINEERING

Section - 1: Civil Engineering (General)

Engineering Surveying, Basic principles of surveying, types and classification of survey, surveying equipment, levelling, Maps and Map making, Indian Map series. Engineering Mechanics, Basic terminology, Units and dimensions, Force, Torque, Friction, Laws of mechanics Free body diagram, solid mechanics. Construction Materials, Types and characteristics of different types of construction materials to include Steel, Concrete, Wood, Stone, Brick/Masonry etc, requirements of good construction materials. Transportation and Highway Engineering – Types and classification of roads, IRC, road construction and materials, road construction equipment. Project Management, Basic terms in project management, , CPM/PERT, critical path, float/slack, quality control, risk management, resource scheduling and management, project management software.

Section - 2 : Mathematics

Linear Algebra - Vector calculus; matrices; similarity transformations, diagonalization, Cayley-Hamilton Theorem, Hermitian, skew Hermitian, unitary matrices, eigenvalues and eigenvectors; linear differential equations: second order linear differential equations; Partial differential equations,

Laplace transform, Fourier series & analysis; Tensors. Numerical Analysis – Numerical Analysis - Numerical solution of algebraic equations using Gauss elimination and Gauss-Siedel methods, numerical solution of ordinary differential equations using Picard, Euler method. Interpolation. Statistics and Probability-mean,mode, median and standard deviation, Probability space, conditional probability, Baye's theorem, various types of distributions, uniform, binomial, poisson, normal and exponential distribution.

Section - 3 : Hydraulics and Water Resource Engineering

Fluid Mechanics: Properties of fluids, fluid statics; Continuity, momentum, and energy equations and their applications; Potential flow, Laminar, and turbulent flow; Flow in pipes, pipe networks; Concept of boundary layer and its growth; Concept of lift and drag. Open Channel Flow and Hydraulic Machines: Introduction of open channel flow, energy depth relationship, uniform flow, gradually varied flow, water surface profiles, and its computations, rapidly varied flow and its computations, spatially varied flow, unsteady flows, hydraulics of mobile bed channels, Fluid machinery; Turbines, pumps and hydro-electric power systems. Water Resources Engineering: Hydrologic cycle and its components, hydrograph analysis, flood estimation, and routing, surface runoff models, hydrological modelling, groundwater hydrology - steady state well hydraulics and aquifers; Application of Darcy's Law, Crop water requirements - Duty, delta, Diversion headworks, canal falls, Regulators modules, Design and construction of gravity dams, Theories of seepage and design of weirs, barrages and dams, spillways, energy dissipators.

Section - 4 : Structural Engineering

Mechanics of Materials: Bending moment and shear force in statically determinate beams; Simple stress and strain relationships; Theories of failures; Simple bending theory, flexural and shear stresses, shear centre; Uniform torsion, buckling of column, combined and direct bending stresses. Structural Analysis: Statically determinate and indeterminate structures by force/ energy methods; Method of superposition; Analysis of trusses, arches, beams, cables and frames; Displacement methods: Slope deflection and moment distribution methods; Influence lines; Stiffness and flexibility methods of structural analysis. Concrete Structures: Working stress, Limit state and ultimate load design concepts; Design of beams, slabs, columns, footing, staircase; Pre-stressed concrete.

Steel Structures: Working stress and Limit state design concepts; Design of tension and compression members, beams and beam-columns, column bases; Connections–simple and eccentric, beam-column connections, plate girders and trusses; Plastic analysis of beams and frames. Pre-stressed Concrete: Pre-stressing systems and end anchorages, losses of pre-stress. Analysis and design of members for flexure,shear,bond and bearings, Cable layouts, Design of Pre-stressed Bridges. Structural Dynamics: Free and Forced vibration analysis of Single and Multi degree of freedom systems. Orthogonality relationships of principal modes, Earthquake forces, nature and magnitude, pseudo-static method of approximate evaluation of earthquake forces, seismicity, Earthquake Motion and Response, Response Spectra, Philosophy of Capacity Design. Concepts of seismic design: Earthquake resistant design, construction & detailing for RCC and Masonry structures and relevant codes such as IS 1893:2002, IS 13920, IS:4326etc.

Section - 5 : Geotechnical Engineering

Soil Mechanics and Geotechnical Engineering- Nature of soil, soil formation and soil type, soil properties, basic definitions, phase relations, index properties, basic concepts of clay minerals and soil structure, soil classification and identification, hydraulic conductivity, seepage, compaction, stress distribution, shear strength, Mohr's circle of stress, Mohr-Coulomb failure theory, shear strength parameters, compressibility and consolidation, soil exploration, shallow foundations, settlements of footings andrafts, pile foundations. Geotechnical Earthquake Engineering - Engineering seismology,seismic risks and hazards, social and economic consequences, nature and attenuation of earthquake magnitude, ground motion, site characteristics, dynamic behavior of soils, liquefaction and cyclic mobility, seismic response of soil structure system, mitigation techniques. Rock Mechanics - Problems of rock mechanics, rock exploration, Griffith's theory, Coulomb's theory, in-situ tests on rock mass, mechanical, thermal and electrical properties of rock mass, pressure tunnels, lined and unlined tunnels, foundation on rocks, slope stability, rock bolt anchors and grouting, problems associated with tunnels, tunnelling in various subsoil conditions and rocks. Uncertainties, Risk and Reliability in Geotechnical Engineering-Risk,randomness, uncertainty, measures of reliability, modeling of uncertainty, probability, tests of goodness-of-fit (chi-square test, Kolmogorov-Smirnov test), reliability methods, deterministic and probabilistic approaches, Monte Carlo simulation and applications, risk assessment. Ground Improvement Techniques-Mechanical modification, precompression, sand drains, soil stabilisation, chemical modifications and grouting, hydraulic modification, ground modification by soil reinforcement, difficult soils.

5. Department of Computer Science and Engineering/ Department of Information Technology / Department of Software Engineering

Computing Related Mathematics: Propositional and first-order logic, sets, relations, functions, partial order and lattice groups. Groups. Vectors, matrices, determinants, System of linear equations, eigenvalues and eigenvectors, LU decomposition. Vector space, differential equation gradients, maxima minima random variables. Uniform Gaussian exponential, Poisson and binomial distributions. Mean Median Mode and standard deviation. Conditional Probability and Bayes Theorem.

Programming and Data Structures: Programming in C, recursion, arrays, stacks, queues, linked list, trees, binary heaps, graphs.

Algorithms: Searching sorting hashing asymptotic worst-case time and space complexity algorithm design techniques: Greedy, dynamic programming and divideand conquer. Graph search, minimum spanning trees, shortest path.

Theory of Computation: Regular expression and finite automata context-free grammar and pushdown automata, regular and context-free languages,pumping Lemma, Turing machines and undecidability.

Compiler Design: Lexical analysis parsing syntax-directed translation runtime environments intermediate code generation

Operating System: Processes inter-process communication concurrency and synchronisation, deadlock CPU scheduling, memory management and virtual memory file systems.

Database Management Systems: ER model, relational algebra, tuple calculus,SQ Lintegrity constraints, normal forms, file organisation, indexing (B and B plus trees) transactions and concurrency control.

Computer Networks: Concept of layering. LAN Technologies (Ethernet) flow and error control techniques, switching ipv4/ ipv5 routers and routing algorithms (distance vector, link state). TCP/UDP and socket congestion control, Application layer protocols (DNS, SMTP POP3 HTTP), Basics of Wi-Fi, network security: Authentication, basics of Public key and Private key cryptography digital signatures and certificates, firewalls.

CIDR notation, Basics of IP support protocols (ARP, DHCP, ICMP), Network Address Translation (NAT), and Email.

Software Engineering: Software development life cycle, requirement and feasibility analysis, data flow diagrams, process specifications, input/ output planning and managing the project, design, coding, software testing, implementation, maintenance, software metrics.

Web Technologies: Web IR retrieval,Web mining, Bigdata, NOSQL, Basics of cloud (SaaS, PaaS, IaaS, Public and Private Cloud).

Artificial Intelligence: Artificial intelligence approach for problem-solving, Automated reasoning for prepositional logic, state-space representation of problems, bounding functions, breadth-first search, depth- first search, A, A*, Ao*, etc. Frames scripts semantic nets, production system, procedural representation.

Basics of Artificial Neural Networks (ANN): Supervised, Unsupervised and Reinforcement Learning.

6. Department of Delhi School of Management

Managerial themes such as E-Governance, Information Technology Management, Strategic Management, Marketing Management, Distribution and Retail Management, Organizational Behavior, Human Resources Management, Corporate Governance and Ethics, Public Policy and Governance, Accounting and Finance,Portfolio Management, Mergers and Acquisition,

Corporate Restructuring, Knowledge Management, General Management Principle and Practices, Supply Chain Management, Business Research Methods, Business Statistics.

7. Department of Design

1. Visualization and Spatial ability
2. Environmental and Social Awareness
3. Analytical and Logical Reasoning
4. Language and Creativity
5. Observation and Design Sensitivity

8. Department of Electrical Engineering/ COE-EVRT

Section 1 : Electric Circuits

Network graph, KCL, KVL, Node and Mesh analysis, Transient response of dc and ac networks, sinusoidal steady-state analysis, resonance, passive filters, Ideal current and voltage sources, Thevenin'stheorem,Norton's theorem, Superposition theorem, Maximum power transfer theorem, Two-port networks, Three phase circuits, Power and power and factor in ac circuits.

Section 2 : Electromagnetic Fields

Coulomb's Law, Electric Field Intensity, Electric FluxDensity,Gauss's Law, Divergence, Electric field and potential due to point, line plane and spherical charge distributions, Effect of dielectric medium, Capacitance of simple configurations, Biot-Savart's law, Ampere's law, Curl, Faraday's law, Lorentz force, Inductance, Magneto motive force, Reluctance, Magnetic circuits, Self and Mutual inductance of simple configurations.

Section 3 : Signals and Systems

Representation of continuous and discrete- time signals, Shifting and scaling operations,Linear Time Invariant and Causal systems, Fourier series representation of continuous periodic signals, sampling theorem, Applications of Fourier Transform, Laplace Transform and z-Transform.

Section 4 : Electrical Machines

Single phase transformer; equivalent circuit, phasor diagram, open circuit and short circuit tests, regulation and efficiency; Three phase transformer; connections, parallel operation; Auto-transformer, Electron mechanical energy conversion principles, DCmachines;

separately excited, series and shunt,motoring and generating mode of operation and their characteristics, starting and speed control of dc motors; Three phase induction motors, Principle of operation, types, performance, torque-speed characteristics, no-load and blocked rotor tests, equivalent circuit, starting and speed control; Operating principle of single phase induction motors; Synchronous machines: cylindrical and salient pole machines, performance, regulation and parallel operation of generators, starting of synchronous motor, characteristics; Types of losses and efficiency calculations of electric machines.

Section 5 : Power systems

Power generation concepts, ac and dc transmission concepts, Models and performance of transmission lines and cables, Series and shunt compensation, Electric field distribution and insulators, Distribution systems, Per-unit quantities, Bus admittance matrix, Gauss-Seidel and Newton-Raphson load flow methods, Voltage and Frequency control, power factor correction, Symmetrical components, Symmetrical and unsymmetrical fault analysis, Principles of over-current,differential and distance protection; Circuit breakers, System stability concepts, Equal area criterion.

Section 6 : Control Systems

Mathematical modeling and representation of systems,Feedback principle, transfer function,Block diagrams and Signals flow graphs, Transient and Steady-state analysis of linear time invariant systems, Routh-Hurwitz and Nyquist criteria, Bode plots Root loci, Stability analysis, Lag, Lead and Lead-Lag compensators; P, PL and PID controllers; State space model, State transition matrix.

Section 7 : Electrical and Electronic Measurements

Bridges and potentiometers, Measurement of voltage, current, power, energy and power factor; Instrument transformers, Digital voltmeters and multimeters, Phase, Time and Frequency measurement; Oscilloscopes, Error analysis.

Section 8 : Analog and Digital Electronics

Characteristics of diodes, BJT, MOSFET; simple diode circuits clipping, clamping, rectifiers; Amplifiers; Biasing, Equivalent circuit and Frequency response; Operational amplifiers; Characteristics and applications, Combinational and Sequential logic circuits, multiplexer, DE multiplexer, 8085 microprocessors; Architecture, programming and Interfacing.

Section 9 : Power Electronics

Characteristics of semiconductor power devices; Diode, Thyristor, Triac, GTO, MOSFET, IGBT; DC to DC conversion; Buck, Boost and Buck-Boost converters; Single and three phase configuration of uncontrolled rectifiers, Line commutated thyristor based converters, Bidirectional ac to dc voltage source converters, Issues of line current harmonics, Power factor, Distortion factor of ac to dc converters, Single phase and three phase inverters, Sinusoidal pulse width modulation.

9. Department of Electronics and Communication Engineering

Networks, Signals and Systems

Network Analysis: Node and mesh Analysis, superposition, Thevenin's theorem and Norton's theorem, maximum power transfer, Steady state sinusoidal analysis: phasors; Time and frequency domain analysis of linear circuits; Solution of network equations using Laplace trans form; Frequency domain analysis of RLC circuits.

Continuous-Time Signals: Fourier series and Fourier transform, sampling theorem and applications.

Discrete-Time Signals: DTFT, DFT, FFT, Z-transform, interpolation of discrete-time signals (FIR, IIR).

LTI systems; definition and properties causality, stability, impulse response, convolution, poles and zeros, parallel and cascade structure, frequency response, group delay, phase delay, digital filter design techniques.

Image Processing

Image Transforms: Unitary transorms, Sine, Cosine, Hadamard transform, KL transform, Short term Fourier transform, Wavelet transform,DWT.

Image Enhancement Techniques- Histogram Modelling, Spatial operations, transforms operations.

Frequency Domain- smoothing frequency domain filters-sharping frequency domain filters, homographic filtering.

Segmentation Techniques- Thresholding based, cluster analysis, region growing.

Morphological Operation- Dilation, Erosion. Histogram and Histogram equalization. Feature Extraction Techniques. Image restoration-Weinerfilter, Image reconstruction radon transform and inverse radon transform.

Electronic Devices

Energy bands in intrinsic and extrinsic semiconductors; Carrier transport; diffusion current, drift current, mobility and resistivity; Generation and recombination of carriers; Poisson and continuity equations; P-N junction, Zener diode, BJT, MOS capacitor, MOSFET, LED, photodiode and solar cell; Integrated circuit fabrication process; oxidation, diffusion, ion implantation, photo lithography, and twin-tub CMOS process.

Analog electronics

Small signal equivalent circuits of diodes, BJTs and MOSFETs; Simple diode circuits; clipping, clamping and rectifiers; Single-stage BJT and MOSFET amplifiers; biasing, bias stability, mid frequency small signal analysis and frequency response; BJT and MOSFET amplifiers; multi-stage, differential, feedback, power and operational; Simple op-amp circuits; Active filters; Sinusoidal oscillators; criterion for oscillation, single-transistor and op-amp configurations; Function generators, wave-shapingcircuitsand555timers;Voltage reference circuits; Power supplies; ripple removal and regulation.

Digital Circuits

Number systems; Combinatorial circuits; Boolean algebra, minimization of functions using Boolean identities and Karnaugh map, logic gates and their static CMOS implementations, arithmetic circuits, code converters, multiplexes, decoders, encoders, PALs and PLAs; Sequential circuits; latches and flip-flops, counters, shift-registers and finite state machines; Data converters; sample and hold circuits, ADCs and DACs; Semiconductor memories; ROM, SRAM, DRAM; 8-bit microprocessor (8085); architecture, programming, memory and I/O interfacing.

Control Systems

Basic control system components; Feedback principle; Transfer function; Block diagram representation; Signal flow graph; Transient and steady-state analysis of LTI systems; Frequency response; Routh-Hurwitz and Nyquist stability criteria; Bade and root-locus plots; Lag, lead and lag-lead compensation; State variable model and solution of state equation of LTI systems.

Communications

Analog Communications: amplitude modulation and demodulation, angle modulation and demodulation, spectra of AM and FM, super heterodyne receivers, circuits for analog communication; Information theory: entropy, mutual information and channel capacity theorem; Digital communications: PCM, DPCM, digital modulation schemes, amplitude, Gram Schmidt orthogonalization procedure, phase and frequency shift keying (ASK, PSK, FSK), QAM, MAP and ML decoding, matched filter receiver, calculation of bandwidth, SNR and BER for digital modulation; Fundamentals of error detection and correction, Single parity code, Hamming codes; Timing and frequency synchronization, inter-symbol interference and its mitigation; Basics of TDMA, FDMA and CDMA.

Probability and Statistics: Mean, mode, median and standard deviation, distribution function, density function, conditional probability, Baye'stheorem uniform, bionomial, Poisson, normal and exponential distribution, Gaussian distribution function, movements, centralized moments, characteristic functions.

Random Process: autocorrelation and power spectral density, properties of white noise, filtering of random signals through LTI system.

Electro magnetics:

Electro magnetics; Maxwell's equations: differential and integral forms and their interpretation, boundary conditions, wave equation, Poynting vector; Plane waves and properties: reflection and refraction, polarization, phase and group velocity, propagation through various media, skindepth; Transmission lines: equations, characteristic impedance, impedance matching, impedance transformation, S-parameters, Smith chart, Rectangular and circular Waveguide, Waveguide components,klystrons, Magnetron,TWT, microwave solid state microwave devices (Gunn, Tunnel, IMPATT diode), dipole and monopole antennas.

10. Department of Environmental Science and Engineering

Characteristics of water and waste water, Water quality requirements, water &waste water treatment, Air quality, Air Pollution &control, Solid waste, Solid Waste management, Engineering system for resource and energy recovery, Industrial waste management, Environmental impact assessment.

11. Department of Humanities

DISCIPLINE : English

Unit 1: History of English Literature

British Literature from Chaucer to the present day: Age of Chaucer, Age of Renaissance, Restoration Period, Neo-Classical Age, Romantic Age, Modern Age, Post-Modern Age.

Unit 2: History of English Language

Evolution of English Language, Rhetoric and Prosody, English Language Teaching

Unit 3: Literature Theory and Criticism

Plato to T.S. Eliot, Marxist Theory, Feminist Theory, Psychoanalytic Theory, Neo-Historicism, Modernism, Structuralism, Post-Modernism, Post-structuralism, Environmentalism

Unit 4: Classical European and Non-British Literature

European Literature from the classical Age to the 20th century, Indian writing in English and Indian Literature in English translation, American and othernon-British English Literature

The Following Syllabus is common for :

DISCIPLINE : Humanities (Economics)

DISCIPLINE : USME (Economics)

Micro Economics

Consumer behaviour, Demand and Supply analysis, Concept of Elasticity. Theory of production and costs, Forms of Market, Pricing and Output Decision. Tax and Subsidy, Elements of General Equilibrium and Welfare Economics.

Macro Economics

Determination of output and Employment, National Income- Concept and Determinants:

Concept of money, bank, Inflation- Causes and Remedies, Concept of multiplier, Business Cycle, IS and LM function. Concept of Growth and Development: Various models of Growth.

International Trade

Theories of International Trade, Balance of Payment, Terms of Trade, Free trade and Protection.

Indian Economy

Main Features: Geographic Size, Natural Resources, Population, Poverty, Agriculture, Industry, Unemployment, Public finance, Meaning and Measurement of Growth Development-meaning and Characteristics of Underdevelopment.

Statistics

Measures of Central tendency, Measurement of Dispersion, Correlation, Regression, Interpolation and Extrapolation Sampling Distributions Normal, t, Chi square, F distribution, Testing of hypothesis, Index numbers.

12. Mechanical Engineering (25% from each section)

Section 1: Applied Mechanics and Design

Engineering Mechanics: Free-body diagrams and equilibrium; friction and its applications including rolling friction, belt-pulley, brakes, clutches, screw jack, wedge, vehicles, etc.; trusses and frames; virtual work; kinematics and dynamics of rigid bodies in plane motion; impulse and momentum (linear and angular) and energy for mulations; Lagrange's equation.

Mechanics of Materials: Stress and strain, elastic constants, Poisson's ratio; Mohr'scircle for plane stress and plane strain; thin cylinders; shear force and bending moment diagrams; bending and shear stresses; concept of shear centre; deflection of beams; torsion of circular shafts; Euler's theory of columns; energy methods; thermal stresses; strain gauges and rosettes; testing of materials with universal testing machine; testing of hardness and impact strength.

Theory of Machines: Displacement, velocity and acceleration analysis of plane mechanisms; dynamic analysis of linkages; cams; gears and gear trains; fly wheels and governors; balancing of reciprocating and rotating masses; gyroscope.

Vibrations: Free and forced vibration of single degree of freedom systems, effect of damping; vibration isolation; resonance; critical speeds of shafts.

Machine Design: Design for static and dynamic loading; failure theories; fatigue strength and the SN diagram; principles of the design of machine elements such as bolted, riveted and welded joints; shafts,gears,rolling and sliding contact bearings, brakes and clutches, springs.

Section 2: Fluid Mechanics and Thermal Sciences

Fluid Mechanics: Fluid properties; fluid statics, forces on submerged bodies, stability of floating bodies; control-volume analysis of mass, momentum and energy; fluid acceleration; differential equations of continuity and momentum; Bernoulli's equation; dimensional analysis; viscous flow of incompressible fluids, boundary layer, elementary turbulent flow, flow through pipes, head losses in pipes, bends and fittings;basics of compressible fluid flow.

Heat-Transfer: Modes of heat transfer; one dimensional heat conduction, resistance concept and electrical analogy, heat transfer through fins; unsteady heat conduction, lumped parameter system, Heisler's charts; thermal boundary layer, dimensionless parameters in free and forced convective heat transfer, heat transfer correlations for flow over flat plates and through pipes, effect of turbulence; heat exchanger performance, LMTD and NTU methods; radiative heat transfer, Stefan-Boltzmann law, Wien's displacement law, black and grey surfaces, view factors, radiation network analysis

Thermodynamics: Thermodynamic systems and processes; proper ties of pure substances, behavior of ideal and real gases; zeroth and first laws of thermodynamics, calculation of work and heat in various processes; second law of thermodynamics; thermodynamic property charts and tables, availability and irreversibility; thermodynamic relations.

Applications: Power Engineering: Air and gas compressors; vapour and gas power cycles, concepts of regeneration and reheat. I.C. Engines: Air-standard Otto, Diesel and dual cycles. Refrigeration and air-conditioning: Vapour and gas refrigeration and heat pump cycles; properties of moist air, psychrometric chart, basic psychrometric processes. Turbo machinery: Impulse and reaction principles, velocity diagrams, Pelton-wheel, Francis and Kaplan turbines; steam and gas turbines.

Section 3: Materials and Manufacturing

Engineering Materials: Structure and properties of engineering materials, phase diagrams, heat treatment, stress-strain diagrams for engineering materials.

Casting, Forming and Joining Processes: Different types of castings, design of patterns, moulds and cores; solidification and cooling; riser and gating design. Plastic deformation and yield criteria; fundamentals of hot and cold working processes; load estimation for bulk (forging, rolling, extrusion, drawing) and sheet (shearing, deep drawing, bending) metal forming processes; principles of powder metallurgy. Principles of welding, brazing, soldering and adhesive bonding.

Machining and Machine Tool Operations: Mechanics of machining; basic machine tools; single and multi-point cutting tools, tool geometry and materials, tool life and wear; economics of machining; principles of non-traditional machining processes; principles of work holding, jigs and fixtures; abrasive machining processes; NC/CNC machines and CNC programming.

Section 4: Industrial Engineering

Metrology and Inspection: Limits, fits and tolerances; linear and angular measurements;comparators; interferometry; form and finish measurement; alignment and testing methods; tolerance analysis in manufacturing and assembly; concepts of coordinate-measuring machine (CMM).

Computer Integrated Manufacturing: Basic concepts of CAD/CAM and their integration tools; additive manufacturing. Robotics and Mechatronics.

Production Planning and Control: Forecasting models, aggregate production planning, scheduling, materials requirement planning; lean manufacturing.

Inventory Control: Deterministic models; safety stock inventory control systems.

Operations Research: Linear programming, simplex method, transportation, assignment, network flow models, simple queuing models, PERT and CPM.

13. USME

Discipline: Management

Managerial themes such as Management Principles,Information Technology Management, Business policy and Strategic Management, Marketing Management, Distribution and Retail Management, Brand and product management, Organizational Behavior and Development, Human Resources Management, Performance and talent management, Corporate Governance and Ethics, Public Policy and Governance, Accounting and Finance, Portfolio Management, Mergers and Acquisition, Corporate Restructuring, Knowledge Management and Practices, Supply chain Management, Business Research Methods, Business Statistics.

Operations management, vendor management, Services marketing and service operations, CRM, Business Analytics, predictive techniques, marketing financial and HR analytics, supply chain analytics, optimization techniques, big data analytics. Entrepreneurship, innovation management, managing technology. Managerial Economics, forecasting, inventory management, various types of costing approaches, total quality management.

Discipline: Innovation, Entrepreneurship and Venture Development

Unit 1: Comprehension Passages from success stories in Entrepreneurship and startup Leadership. Current affairs related to start-ups.

Unit 2: Entrepreneurship, Social entrepreneurship, Intrapreneurship, Entrepreneurial Characteristics, Entrepreneur v/s Manager, Entrepreneurial Motivation.

Unit 3 : Innovation Management, Diffusion of Innovation, Innovation Cycle, New Product Development, Design Thinking.

Unit 4 : Startup, Stages of Startup, Ideation, Pre-startup, Startup and Scaling up Stage, Start Up India Program, Start Up concept and its support to start Ups, Startup Financing, Seed Funding, Angel investors, venture capital funding, government financial support for startups, MSME-definition and its types.

Unit 5: Government Schemes &Policy for Startups and MSMEs, ATAL Innovation Mission (AIM), ATAL Tinkering Lab(ATL),Self Employment &Talent Utilization (SETU) scheme,SFURTI Scheme, Intellectual Property Rights (IPR), Start Up Intellectual Property Protection (SIPP), Trademark, Copyright, Patents, Type of Companies in India and their Characteristics, Formation Legalities of the Company.

14. Centre of Excellence for the Science of Happiness (CESH)

Science and practice of happiness, Measurement of Happiness and well being, Happiness at work and daily living, consumer happiness, building resilience, Gross National Happiness, Traditional knowledge and Values, Theories of emotions, Happiness and prosperity, Harmony in family and society, Human values, Science and art behind meditation, Power of Self discipline, Peaceful conversation, Brain and heart connection, Healing from the heart, Managing with heart.

15. Department of Geospatial Science and Technology (DGST)

Engineering Mathematics:- Surveying measurements, Accuracy, Precision, Most probable value, Errors and their adjustments, Regression analysis, Correlation coefficient, Least square adjustment, Statistical significant value, Chi square test. Remote Sensing: Basic concept, Electromagnetic spectrum, Spectral signature, Resolutions- Spectral. Spatial, Temporal and Radiometric, Platforms and Sensors, Remote Sensing Data Products - PAN, Multispectral, Microwave, Thermal, Hyperspectral, Visual and digital interpretation methods GNSS: Principle used, Components of GNSS, Data collection methods, DGPS, Errors in observations and corrections. GIS:

Introduction, Data Sources, Data Models and Data Structures, Algorithms, DBMS, Creation of Databases (spatial and non-spatial), Spatial analysis - Interpolation, Buffer, Overlay, Terrain Modeling and Network analysis.

Surveying and Mapping. Maps: Importance of maps to engineering projects, Types of maps, Scales and uses, Plotting accuracy, Map sheet numbering, Coordinate systems- Cartesian and geographical, map projections, map datum– MSL, Geoid, spheroid, WGS-84. Land Surveying: Various Levels, Levelling methods, Compass, Theodolite and Total Station and their uses, Tachometer, Trigonometric levelling, Traversing, Triangulation and Trilateration. Aerial Photogrammetry: Types of photographs, Flying height and scale, Relief (height) displacement, Stereoscopy, 3-D Model, Height determination using Parallax Bar, Digital Elevation Model (DEM), Slope.

Image Processing and Analysis Data Quantization and Processing: Sampling and quantization theory, Principle of Linear System, Convolution, Continuous and Discrete Fourier Transform. Digital Image Processing: Digital image characteristics: image histogram and scatterogram and their significance, Variance-Covariance matrix, Correlation matrix and their significance. Radiometric and Geometric Corrections: Registration and Resampling techniques. Image Enhancement: Contrast Enhancement: Linear and Non-linear methods; Spatial Enhancement: Noise and Spatial filters . Image Transformation: Principal Component Analysis (PCA), Discriminant Analysis, Colour transformations (RGB - IHS, CMYK), Indices (Ratios, NDVI, NDWI). Image Segmentation and Classification: Simple techniques.

16. Vinod Dham Centre of Excellence for Semiconductors and Microelectronics (VDSemiX)

Discipline: Semiconductors and Microelectronics

Section 1: Computational Physics

Roots of equations, Bisection method, Regula Falsi Method,Secant, Newton-Raphsonmethod, Convergence, Interpolation: Finite differences and difference operators, Interpolation with equally spaced data points: Newton's forward and backward formulae for interpolation, Central difference: Gauss forward, Gauss Backward, Stirling, Bessels, Everett's formula for interpolation, Numerical solution of ordinary differential equations: Taylor's series method, Picard's method of successive approximation methods, Euler's method, modified Euler's method, Runge-Kutta method, solution of second order and simultaneous differential equations, binomial, poisson, normal and exponential distribution, Application of optimization and variational methods to problem of interest in semiconductors and microelectronics.

Section 2: Semiconductor Device Physics

Fermi energy, density of states function, quantum tunneling, potential inside a semiconductor, semiconductor in Equilibrium, Carrier transport phenomena and models, PN junction, metal semiconductor and hetero junctions, Shallow and deep levels, Carrier statistics, Carrier transport, Carrier mobility, Scattering mechanisms, Hall effect measurements, High field property, Non-equilibrium conditions, Quasi Fermi levels, Recombination processes, Current density and continuity equations, Surface recombination, Surface states, Excitons in semiconductors, Diffusion capacitance, Generation-recombination currents, Junction breakdown mechanisms. Hetero junctions: Band alignments, energy band diagrams of hetero junctions, formation of two dimensional electron gas; Metal-semiconductor contacts: Schottky barrier diode, Fermi level pinning,C-V characteristics of a Schottky diode, Current transport processes and I-V characteristics, Ohmic contacts.

Section 3: Electromagnetics

Electromagnetics; Maxwell's equations: differential and integral for ms and the irinterpretation, boundary conditions, wave equation, Poynting vector; Plane waves and properties: reflection and refraction, polarization, phase and group velocity, propagation through various media, skin depth; Transmission lines: equations, characteristic impedance, impedance matching, impedance transformation, S-parameters, Smith chart; Waveguides: modes, boundary conditions, cut-off frequencies, dispersion relations.

Section 4: Electronic Devices

Energy bands in intrinsic and extrinsic silicon; Carrier transport; diffusion current, drift current, mobility and resistivity; Generation and recombination of carriers;Poisson and continuity equations;P-N junction, Zener diode, BJT, MOS MOSFET, LED, photo diode and solar cell, Integrated circuit fabrication process: epitaxy, clean room, oxidation, diffusion, ion implantation, photo lithography, assembly techniques, packaging and twin-tub CMOS process.

Section 5: Analog System Design

Small signal equivalent circuits of diodes, BJTs and MOSFETs; Simple diode circuits; clipping, clamping and rectifiers; Single-stage BJT and MOSFET amplifiers; biasing, bias stability, mid frequency small signal analysis and frequency response; BJT and MOSFET amplifiers; multi-stage, differential, feedback, power and operational; Simple op-amp circuits; Active filters; Sinusoidal oscillators; criterion for oscillation, single-transistor and op-amp configurations; Current Mirror, Voltage and Current references, Single stage amplifiers, Cascode amplifier, Differential amplifiers, Output amplifiers, CMOS Analog opamps, Single stage and two stage Comparators, Sample and Hold circuits, Nyquist rate ADCs and DACs. Function generators, wave-shaping circuits and 555 timers; Voltage reference circuits; Power supplies; ripple removal and regulation

Section 6: Digital System Design

Number systems; Combinatorial circuits; Boolean algebra, minimization offunctionsusing Boolean identities and Karnaugh map, logic gates, arithmetic circuits, code converters, multiplexes, decoders, encoders, PALs and PLAs; Sequential circuits; latches and flip-flops, counters, shift-registers and finite state machines; Data converters; sample and hold circuits, ADCs and DACs; Semiconductor memories; ROM, SRAM, DRAM, Semiconductor Memory Design. Data Path subsystems– Adders, Multipliers, Shifters, Introduction to RTL, DTL TTL, Schottky TTL, IIL and ECL logic family, concept of Noise margin, fan out and propagation delay, 16-bit microprocessor (8086); architecture, programming, memory and I/O interfacing.

Section 7: VLSI System Design

Static and dynamic characteristics of CMOS inverters, delay estimation, logical efforts and transistor sizing, power dissipation, interconnect, combinational CMOS logic circuits, complex logic circuits, behavior of MOS logic elements, SR latch circuit, clocked latch and flip-flop circuits, CMOS D-latch and edge-triggered flip-flop, pass transistor and Transmission gate logic, Stick diagram and Layout, VLSI Design Flow: Full custom and Semi Custom, Verilog/VHDL, FPGA architecture,Verification, and Testing.

AC Model of MOS, MOS resistors, current sink/ source, CMOS inverters - calculation of VTC critical points, power dissipation, rise time and fall time, Fabrication process flow, layout design rules, full custom mask layout design, stick diagram.

Section 8: Semiconductor Optoelectronics

LEDs, LASERs, Semiconductor Photo detectors, Rate of absorption in a semiconductor, materials, device structure, characteristics, power output, responsivity, speed of photo detector, PIN photo diode, APDs, Noise in Photo detectors, Applications, LIGHT MODULATORS AND PHOTOVOLTAIC DEVICES-Basic Principles, Electro-absorption Modulator: Device structure and Characteristics, Solar Cells: Device Physics, I-V characteristics in dark and light, Solar Energy Conversion, Conversion Efficiency, Materials, Applications

17. Department of Biotechnology

Molecules and their Interaction Relevant to Biology: Structure and functions of biomolecules; Carbohydrates; Fatty acids; Lipids; Amino acids; Proteins; Nucleic acids – DNA, mRNA, tRNA, rRNA; Hormones;Vitamins; Enzymes; Bio energetics; Cell metabolism; Protein-protein and protein- DNA interactions

Cellular Organization: Cell theory; Cell as basic unit of life; Hierarchy of cell organization;

Structure and organization of prokaryotic and eukaryotic cells; Structure and function of cell organelles; cell cycle; Bio-membranes; Cytoskeletal elements; Chromosome structure; Karyo type; Chromatin organization

Fundamental Processes: Photosynthesis; Cellular respiration; Movement through cell membrane; Nutrition; Blood clotting; Human physiological systems; Replication; Transcription; Translation; DNA repair mechanisms; Plant physiology; Bacterial growth; Microbial genetics, Secondary metabolites

Cell Communication and Cell Signaling: Tight adherents and communicating cell junctions; Cell adhesion molecules; Cadherins and Integrins; Extracellular matrix; Cell cycle; Basics of cancer; Basics of cell signaling; Major signaling pathways.

Developmental Biology: Stages of development; Mechanism of differentiation; Germ layers; Potency; Morphogenetic movements; Early and late development in model organisms; Cell division; Gametogenesis and fertilization in animals and flowering plants; Embryology; Seed germination; Dormancy; Evolution and natural selection; Mendel's law of heredity; Evidence of DNA as genetic information carrier; Hardy-Weinberglaw; Extra-chromosomal inheritance; Sex- linked inheritance in humans;Mutations

Plant and Animal Biotechnology: Plant tissue culture techniques; Totipotency; Organogenesis and Somatic embryo genesis; Suspension culture; Protoplast isolation and somatic hybridization; Production of secondary metabolites; Basic techniques in animal cell and organ culture; Bioreactors for large scale culture of animal cells; Stem cells; Transgenic plants and animals

Immunology and Immunotechnology: Immunity; Antigen; Structure of antibody; Hapten; Antigen-antibody interaction, Introduction to antigen presentation; Role of MHC; Complement system; Bacterial diseases of humans; Types of vaccines; Immunization; Recombinant vaccines.

Inheritance Biology: Mendelian's principles, Extensions of Mendelian principles, Gene mapping methods, Extra chromosomal inheritance, Human genetics, Mutations, Structural and numerical alterations of chromosomes.

Diversity of Life Forms: General characteristics of life forms; General characteristics of bacteria, fungi, algae, Microbial growth curve; plant and animal viruses; Classification of plant and animal kingdom

Ecological Principles and Environmental Biology: Ecosystem; Ecological relationships; Habitat and niche; Ecology of eco systems; Air, water, and soil pollution; Greenhouse effect and global warming; Noise pollution;Pollution abatement; Wastewater treatment; Disposal of solid wastes; Bio geochemical cycles of elements; Bio remediation; Bio leaching; Bio pesticides; Bio fertilizers

Evolution and Behaviour: Evolution and natural selection; Mendel's law of heredity; Evidences of DNA as genetic information carrier; Hardy-Weinberg law; Extra-chromosomal inheritance; Sex-linked inheritance in humans; Mutations.

Applied Biology: Basics of fermentation technology; Microbes in industry; Biosensors; Biofuels; Principles of gene cloning; Methods of gene transfer; Application of biology in agriculture, health, industry, and environment sectors.

Assay techniques in Biology: Basics of Centrifugation; Electrophoresis; Chromatography; Microscopy; UV- Visible spectrophotometry; Radiotracer technique; PCR; DNA sequencing; Southern blotting; Tests of significance; Analysis of variation; Correlation and regression; Hybridoma technology; Basic techniques in bio informatics

Bio informatics and Computational Biology: Sequence and Structural Databases (NCBI, GenBank, EMBL, DDBJ, PDB); SNP databases; Visualization tools- Pymol, VMD; Functional Annotation; Local and Global Alignment; Phylogenetics; Pharmacogenomics; Machine learning

Diagnostic techniques: X-Rays, CT scan, MRI, Pathology test, ECG, EEG.

18. Centre for Community Development and Research (CCDR)

Discipline: Community Development and Research

- **Theories of Development:**

  Sustainable development, rural development models, human development approach, and community empowerment.

  Principles and skills: Leadership development, building partnerships, and involving NGOs/CBOs.

  Rural and Urban development: Infrastructure, migration, and slum redevelopment.

- **Social & Economic Context:**

  Poverty, inequality, and human development.

  Gender dynamics, social exclusion, and marginalized communities.

  Agrarian change, rural economy, and self-help groups (SHGs).

- **Policy & Governance:**

  Indian planning process and constitutional provisions (Directive Principles).

  Decentralization: Panchayati Raj institutions and local governance.

  Sustainable development goals (SDGs) and social policy formulation.

  Make in India policy. World strategic issues.

- **Environmental & Emerging Issues:**

  Climate change impact, disaster management, and sustainable technologies.

  Corporate Social Responsibility (CSR) and its role in development.

  Quantitative/Qualitative Techniques: Basic social statistics, development indices, and qualitative data analysis techniques.

- **Research Fundamentals**

  Types of research (qualitative, quantitative, applied, analytical).

  Research design, hypothesis formulation, and literature review.

  Data collection methods (survey, interview, case study).

  Sampling techniques (probability & non-probability).

  Statistical tools: mean, median, mode, ANOVA, t-test, chi-square, and correlation/regression.

  Research ethics, plagiarism, report writing, referencing.

  Project proposal preparation.

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